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TAKEDA et al.

(54) **OPERATION DEVICE MANAGEMENT SYSTEM, OPERATION DEVICE MANAGEMENT APPARATUS, OPERATION DEVICE MANAGEMENT METHOD, AND RECORDING MEDIUM**

(71) Applicant: **Mitsubishi Electric Corporation**, Tokyo (JP)

(72) Inventors: **Yuji TAKEDA**, Tokyo (JP); **Naoki KIZU**, Tokyo (JP); **Shinya SUGINO**, Tokyo (JP)

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(57) **ABSTRACT**

An operation device management server includes a control record storage to store, in a manner associated with one another, specific control information indicating specific control performed on a control target device through an operation performed previously on an operation device, operation device ID name information, and control target device ID name information, a search condition acquirer to acquire search condition information indicating a search condition for searching for at least one of control target device ID name information of a control target device, operation device ID name information of an operation device, or specific control information, a searcher to search for at least one of the control target device ID name information, the operation device ID name information, or the specific control information based on the search condition information, and a search result notifier to generate search result information to a terminal.

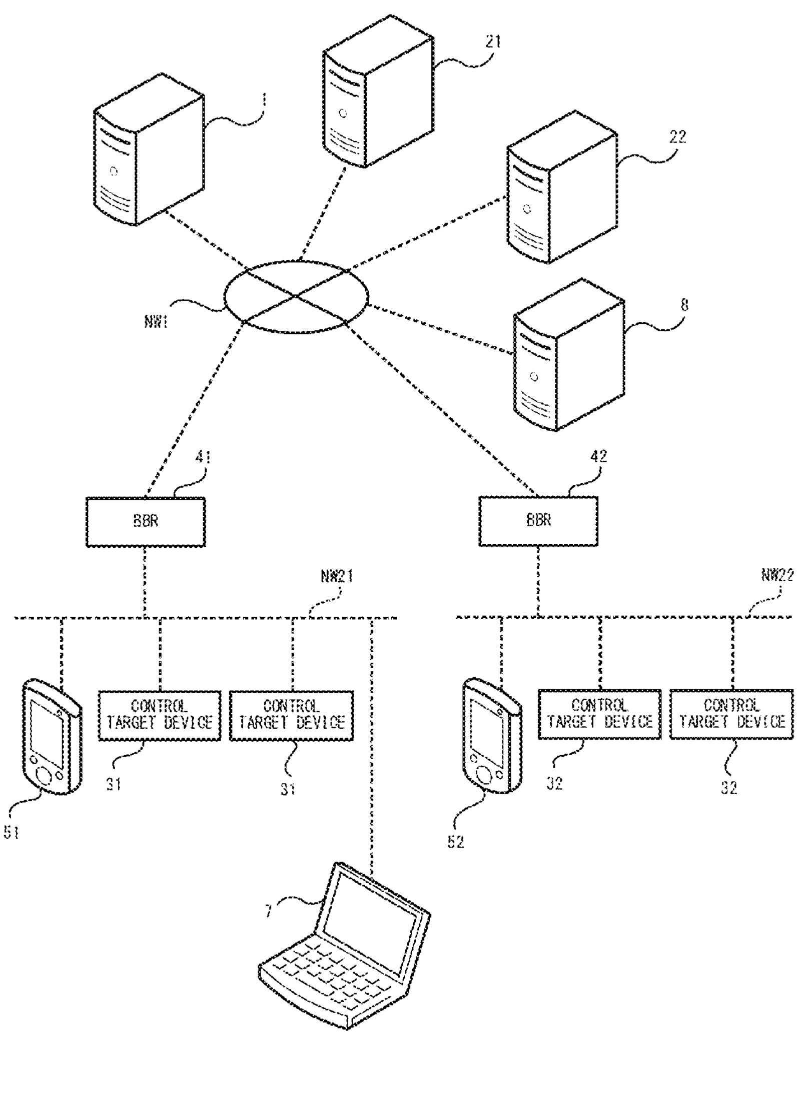
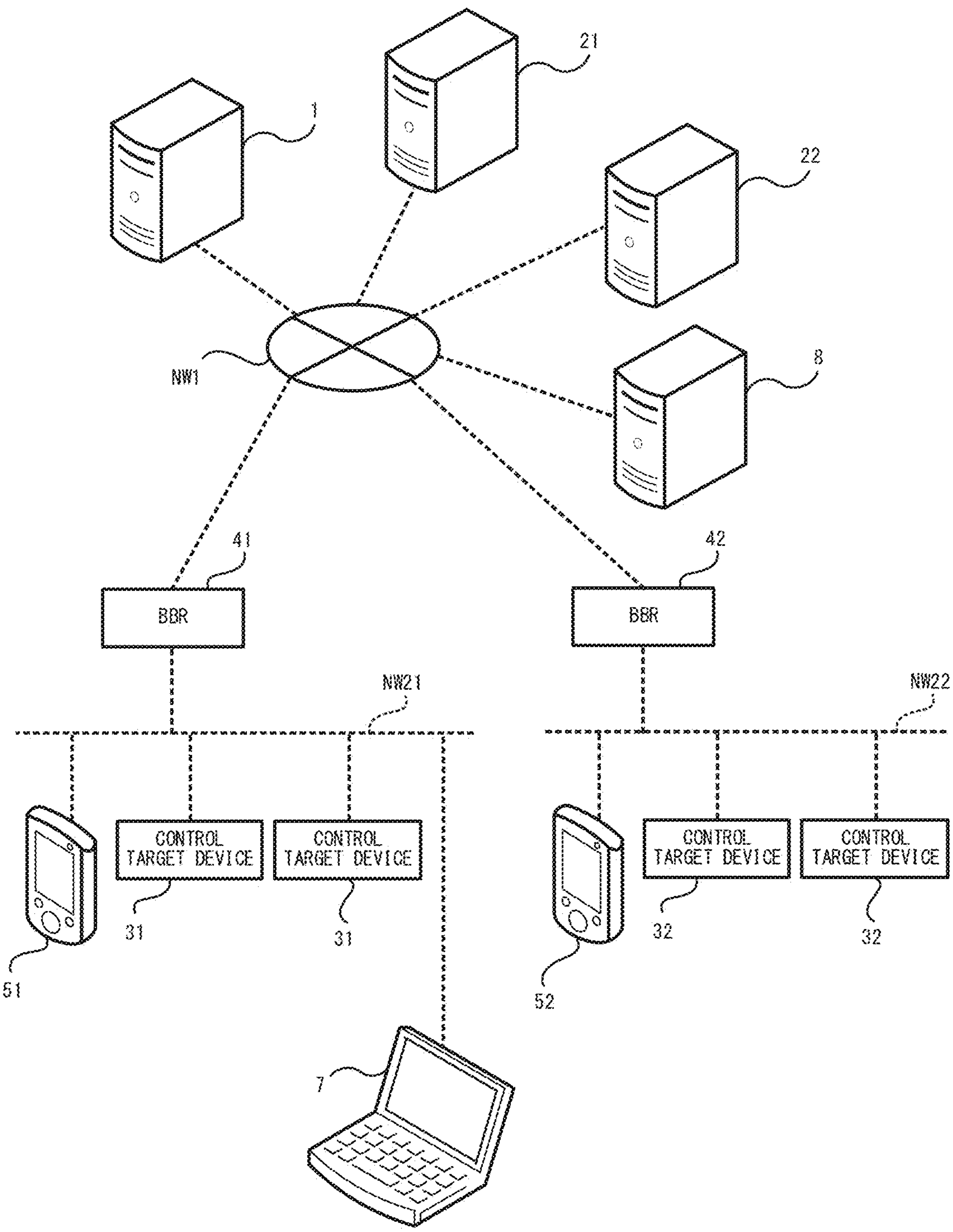


FIG.1



21
1
22
NW1
8
41
42
BBR
BBR
NW21
NW22
CONTROL TARGET DEVICE
CONTROL TARGET DEVICE
CONTROL TARGET DEVICE
CONTROL TARGET DEVICE
31
31
32
32
51
52
7

FIG.2

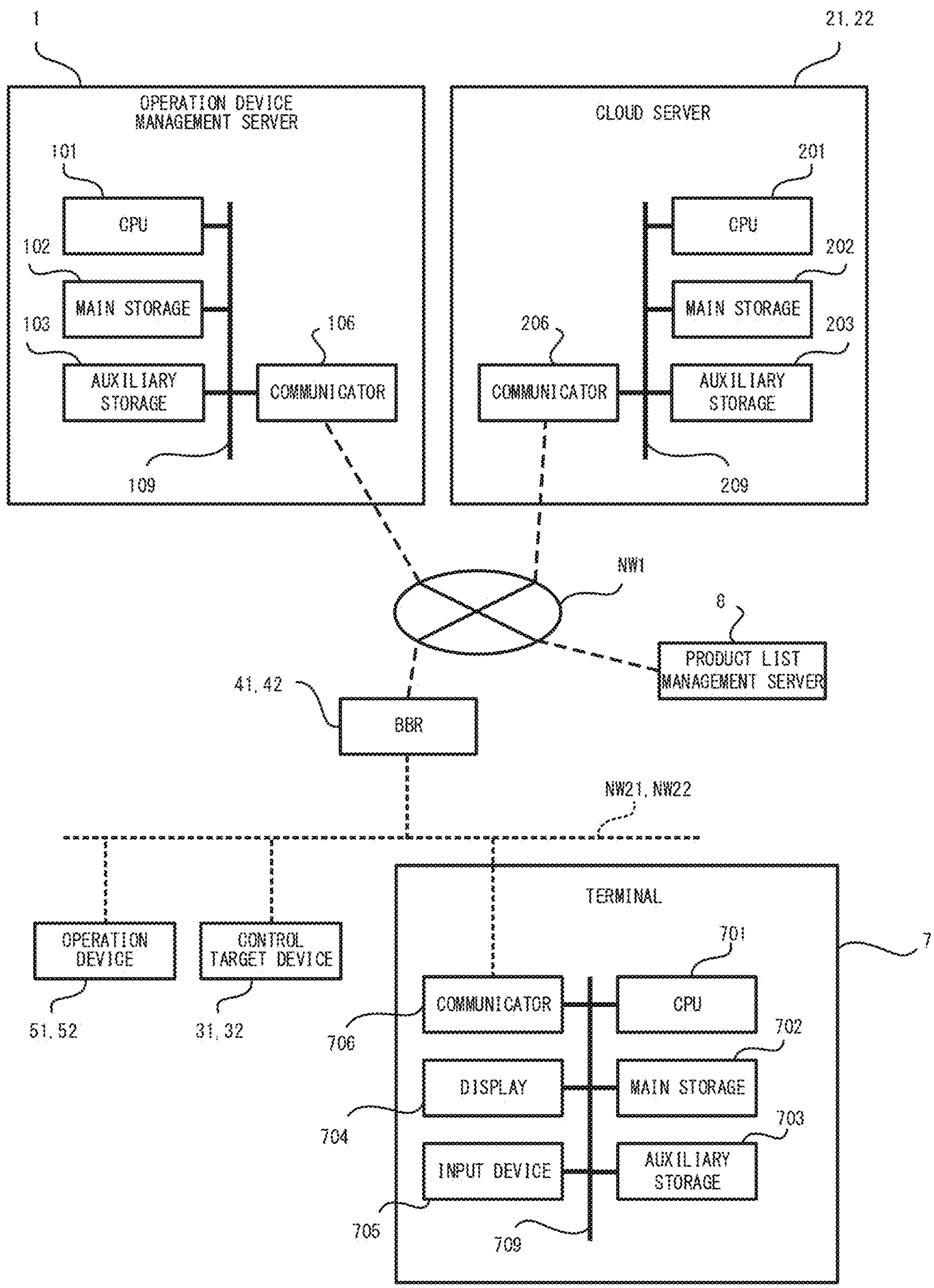

1
OPERATION DEVICE MANAGEMENT SERVER
101
CPU
102
MAIN STORAGE
103
AUXILIARY STORAGE
106
COMMUNICATOR
109
21, 22
CLOUD SERVER
201
CPU
202
MAIN STORAGE
203
AUXILIARY STORAGE
206
COMMUNICATOR
209
NW1
8
PRODUCT LIST MANAGEMENT SERVER
41, 42
BBR
NW21, NW22
OPERATION DEVICE
51, 52
CONTROL TARGET DEVICE
31, 32
TERMINAL
7
701
CPU
702
MAIN STORAGE
703
AUXILIARY STORAGE
706
COMMUNICATOR
704
DISPLAY
705
INPUT DEVICE
709

FIG.3

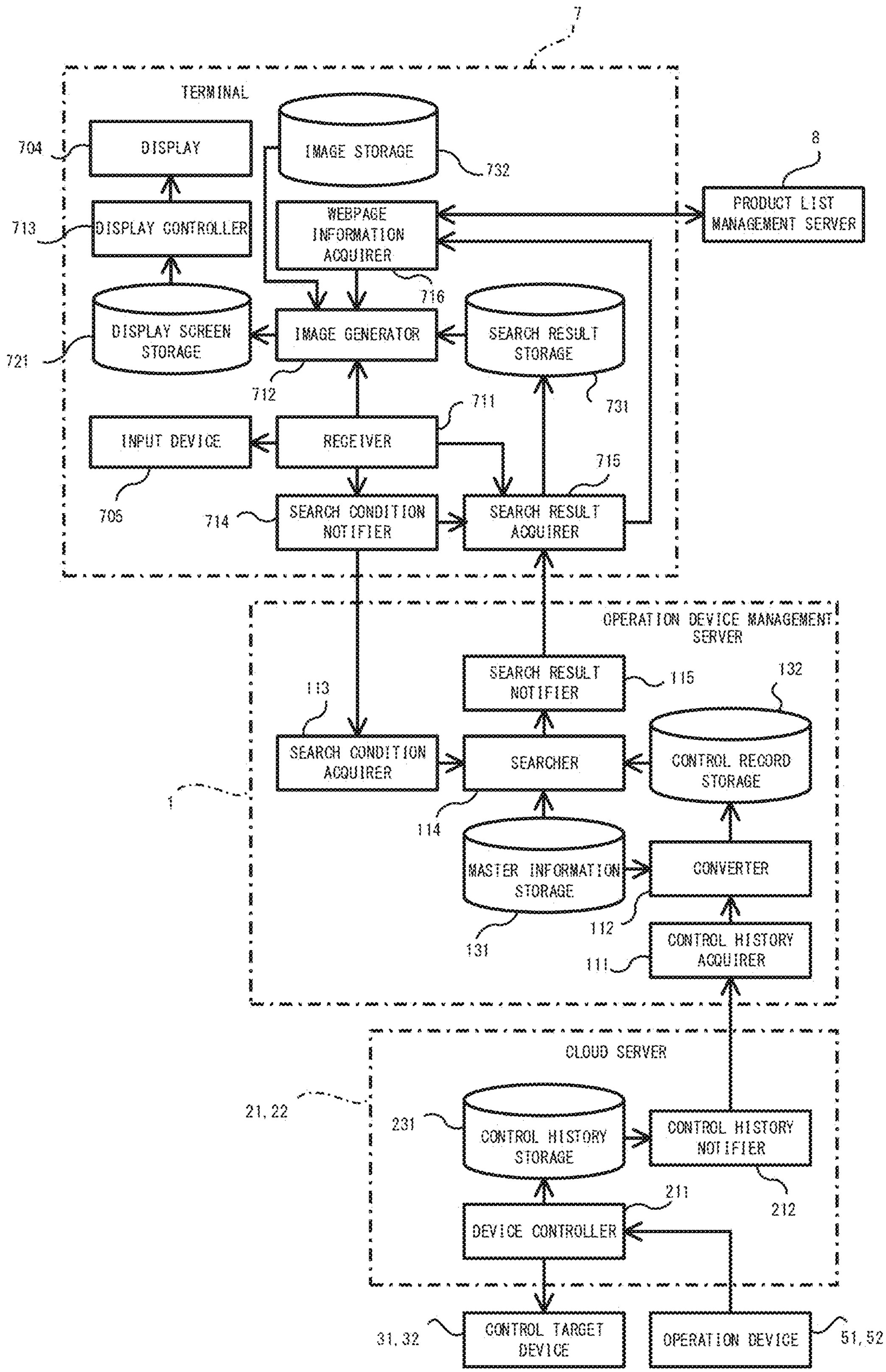

7
TERMINAL
704
DISPLAY
IMAGE STORAGE
732
8
PRODUCT LIST MANAGEMENT SERVER
713
DISPLAY CONTROLLER
WEBPAGE INFORMATION ACQUIRER
716
DISPLAY SCREEN STORAGE
721
IMAGE GENERATOR
712
SEARCH RESULT STORAGE
731
711
INPUT DEVICE
RECEIVER
706
715
714
SEARCH CONDITION NOTIFIER
SEARCH RESULT ACQUIRER
OPERATION DEVICE MANAGEMENT SERVER
SEARCH RESULT NOTIFIER
115
132
113
SEARCH CONDITION ACQUIRER
SEARCHER
CONTROL RECORD STORAGE
1
114
MASTER INFORMATION STORAGE
CONVERTER
112
131
CONTROL HISTORY ACQUIRER
111
CLOUD SERVER
21, 22
231
CONTROL HISTORY STORAGE
CONTROL HISTORY NOTIFIER
212
211
DEVICE CONTROLLER
31, 32
CONTROL TARGET DEVICE
OPERATION DEVICE
51, 52

FIG.4

MASTER INFORMATION STORAGE131

MANUFACTURER CODE	MANUFACTURER NAME
11223344	COMPANY A
12345678	COMPANY B
98765432	COMPANY C
⋮	⋮

CONTROL TARGET DEVICE ID CODE	CONTROL TARGET DEVICE ID NAME	CLASS GROUP	CLASS
AABBCCDDEEFFG	AirCON-X	0x01	0x30
AABBCCDDHIJKLM	AirCON-Y	0x01	0x30
⋮	⋮	⋮	⋮

OPERATION DEVICE ID CODE	OPERATION DEVICE ID NAME	CLASS GROUP	CLASS
AABBCCDDOPQRS	CONTROLLER A	0x05	0xFF
1111	CONTROL APP	0x05	0xFF
2222	EASY CONTROL APP	0x05	0xFF
3333	MR. CONTROL	0x05	0xFF
⋮	⋮	⋮	⋮

SPECIFIC CONTROL CODE	SPECIFIC CONTROL
0x80	POWER ON-OFF SETTING
0xB0	OPERATION MODE SETTING
0xB1	AUTOMATIC TEMPERATURE SETTING
⋮	⋮

FIG.5

CONTROL RECORD STORAGE232

OPERATION DEVICE ID NAME	CLASS GROUP	CLASS	SPECIFIC CONTROL	CONTROL TARGET DEVICE ID NAME	CLASS GROUP	CLASS
MR. CONTROL	0x05	0xFF	OPERATION MODE SETTING	AirCON-X	0x01	0x30
CONTROL APP	0x05	0xFF	POWER ON-OFF SETTING	AirCON-X	0x01	0x30
MR. CONTROL	0x05	0xFF	POWER ON-OFF SETTING	AirCON-Y	0x01	0x30
EASY CONTROL APP	0x05	0xFF	AUTOMATIC TEMPERATURE SETTING	AirCON-X	0x01	0x30
MR. CONTROL	0x05	0xFF	POWER ON-OFF SETTING	AirCON-X	0x01	0x30
MR. CONTROL	0x05	0xFF	OPERATION MODE SETTING	AirCON-Y	0x01	0x30
MR. CONTROL	0x05	0xFF	AUTOMATIC TEMPERATURE SETTING	AirCON-X	0x01	0x30
EASY CONTROL APP	0x05	0xFF	POWER ON-OFF SETTING	AirCON-X	0x01	0x30
...	...	...	...	...	...	...

FIG.6

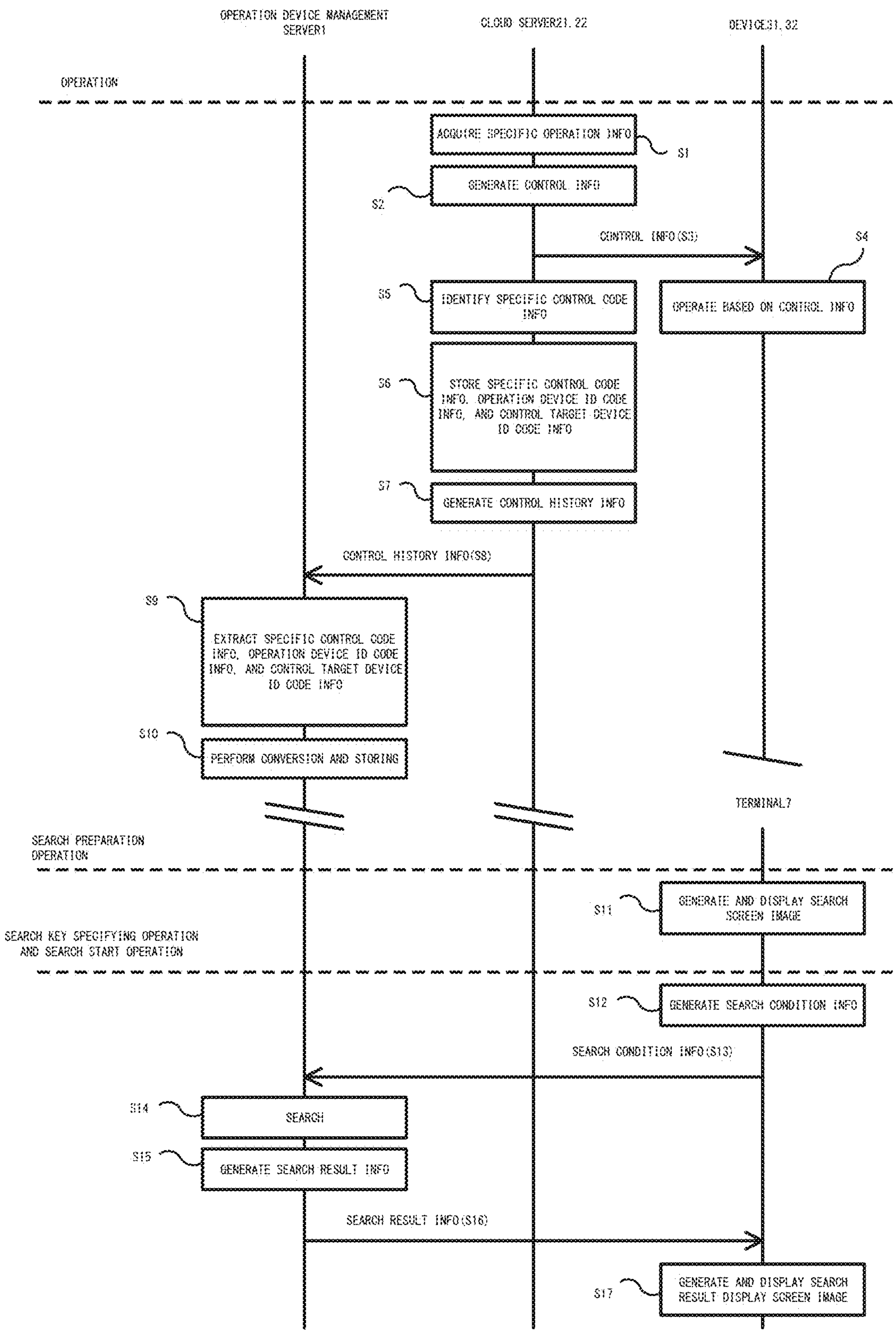

OPERATION DEVICE MANAGEMENT SERVER1
CLOUD SERVER21, 22
DEVICE31, 32
OPERATION
ACQUIRE SPECIFIC OPERATION INFO
S1
GENERATE CONTROL INFO
S2
CONTROL INFO(S3)
S4
S5
IDENTIFY SPECIFIC CONTROL CODE INFO
OPERATE BASED ON CONTROL INFO
S6
STORE SPECIFIC CONTROL CODE INFO, OPERATION DEVICE ID CODE INFO, AND CONTROL TARGET DEVICE ID CODE INFO
S7
GENERATE CONTROL HISTORY INFO
CONTROL HISTORY INFO(S8)
S9
EXTRACT SPECIFIC CONTROL CODE INFO, OPERATION DEVICE ID CODE INFO, AND CONTROL TARGET DEVICE ID CODE INFO
S10
PERFORM CONVERSION AND STORING
TERMINAL7
SEARCH PREPARATION OPERATION
S11
GENERATE AND DISPLAY SEARCH SCREEN IMAGE
SEARCH KEY SPECIFYING OPERATION AND SEARCH START OPERATION
S12
GENERATE SEARCH CONDITION INFO
SEARCH CONDITION INFO(S13)
S14
SEARCH
S15
GENERATE SEARCH RESULT INFO
SEARCH RESULT INFO(S16)
S17
GENERATE AND DISPLAY SEARCH RESULT DISPLAY SCREEN IMAGE

FIG.7

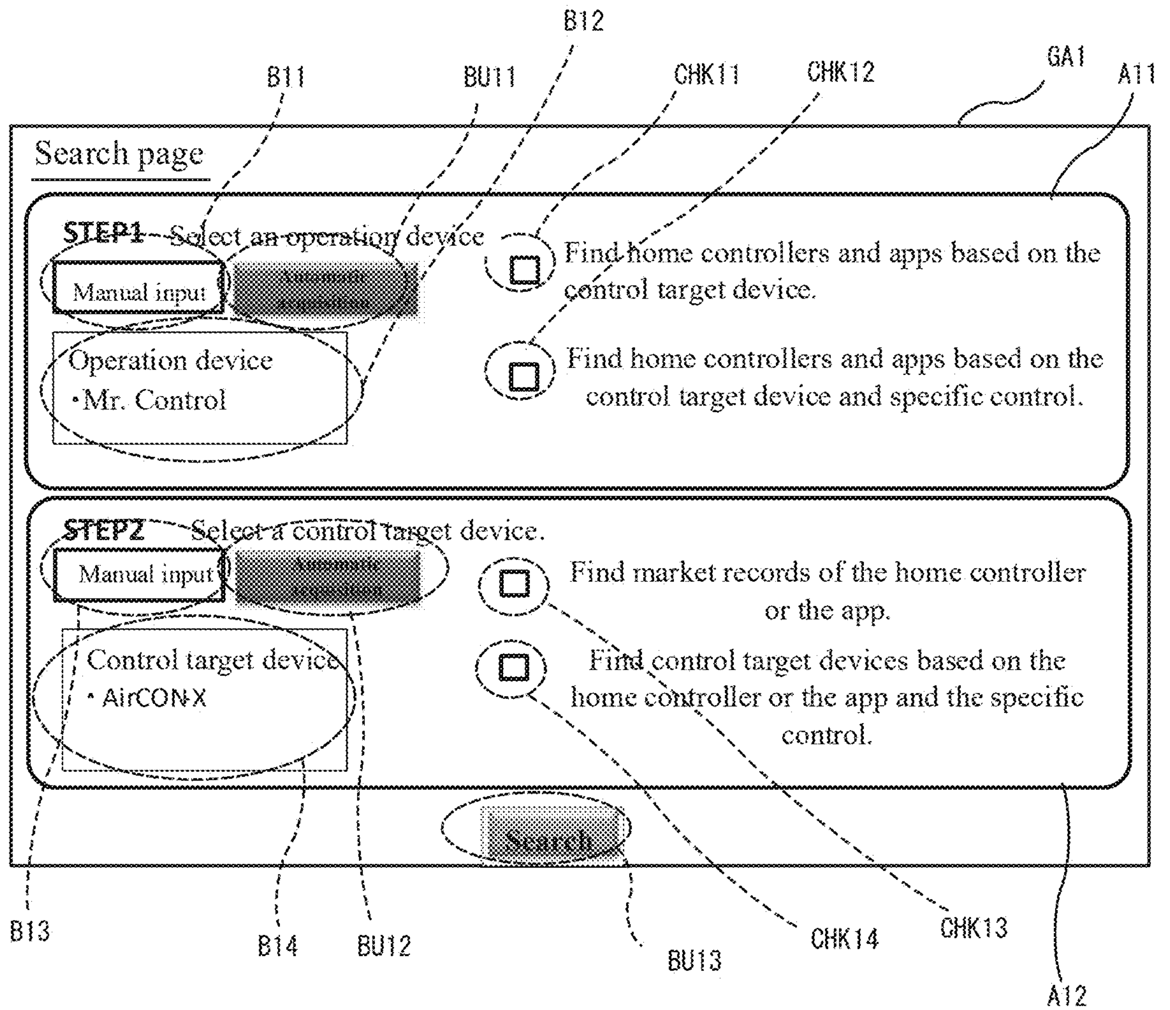

B11
BU11
B12
CHK11
CHK12
GA1
A11
Search page
STEP1 Select an operation device
Manual input
Operation device
•Mr. Control
Find home controllers and apps based on the control target device.
Find home controllers and apps based on the control target device and specific control.
STEP2 Select a control target device.
Manual input
Control target device
• AirCON-X
Find market records of the home controller or the app.
Find control target devices based on the home controller or the app and the specific control.
Search
B13
B14
BU12
BU13
CHK14
CHK13
A12

FIG.8A

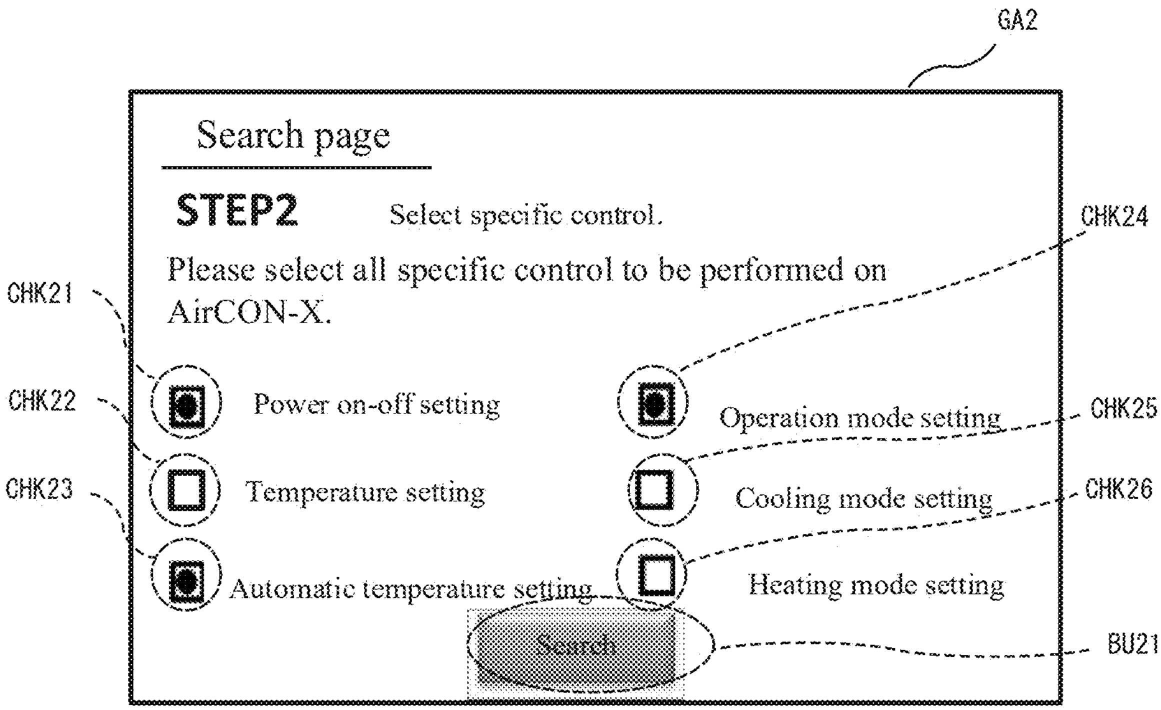

GA2
Search page
STEP2
Select specific control.
Please select all specific control to be performed on AirCON-X.
CHK21
Power on-off setting
CHK22
Temperature setting
CHK23
Automatic temperature setting
CHK24
Operation mode setting
CHK25
Cooling mode setting
CHK26
Heating mode setting
Search
BU21

FIG.8B

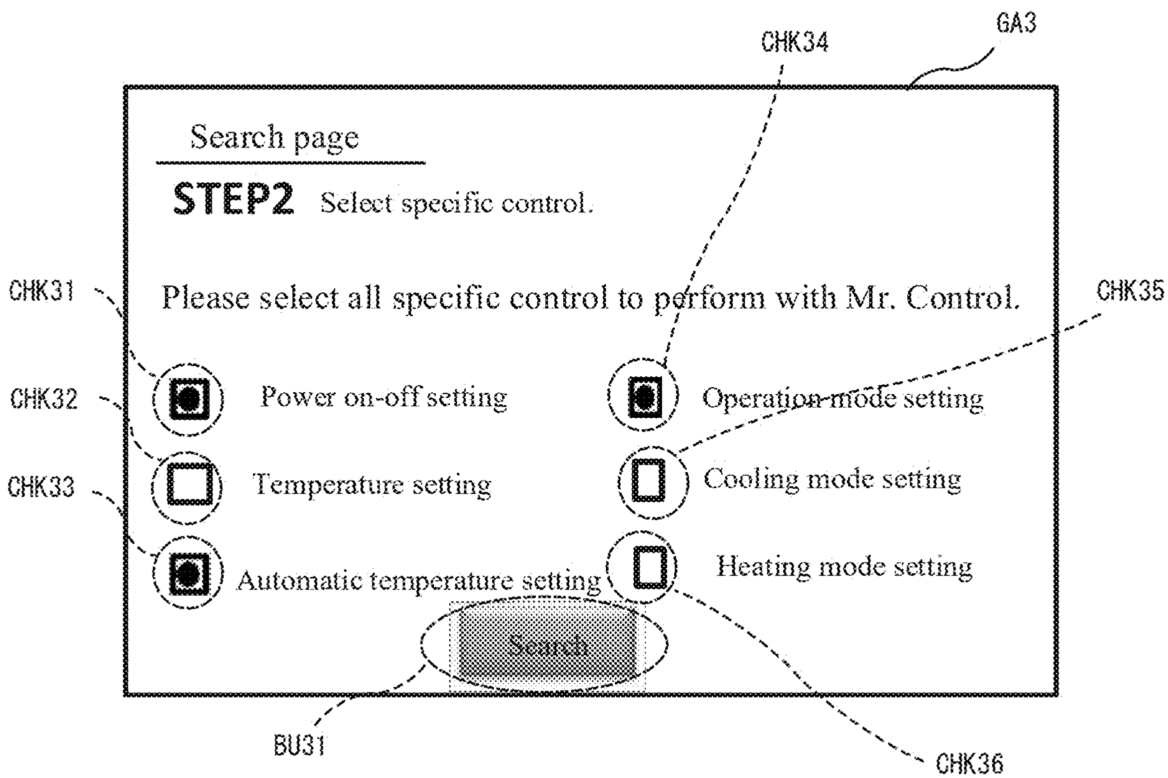

GA3
CHK34
Search page
STEP2 Select specific control.
Please select all specific control to perform with Mr. Control.
CHK31
CHK35
CHK32
CHK33
Power on-off setting
Operation mode setting
Temperature setting
Cooling mode setting
Automatic temperature setting
Heating mode setting
Search
BU31
CHK36

FIG.9A

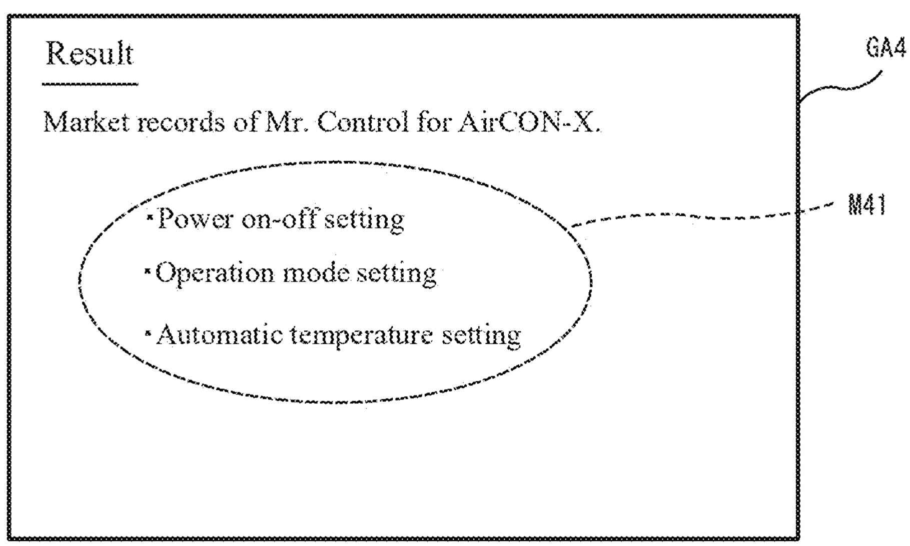

Result
Market records of Mr. Control for AirCON-X.
・Power on-off setting
・Operation mode setting
・Automatic temperature setting
GA4
M41

FIG.9B

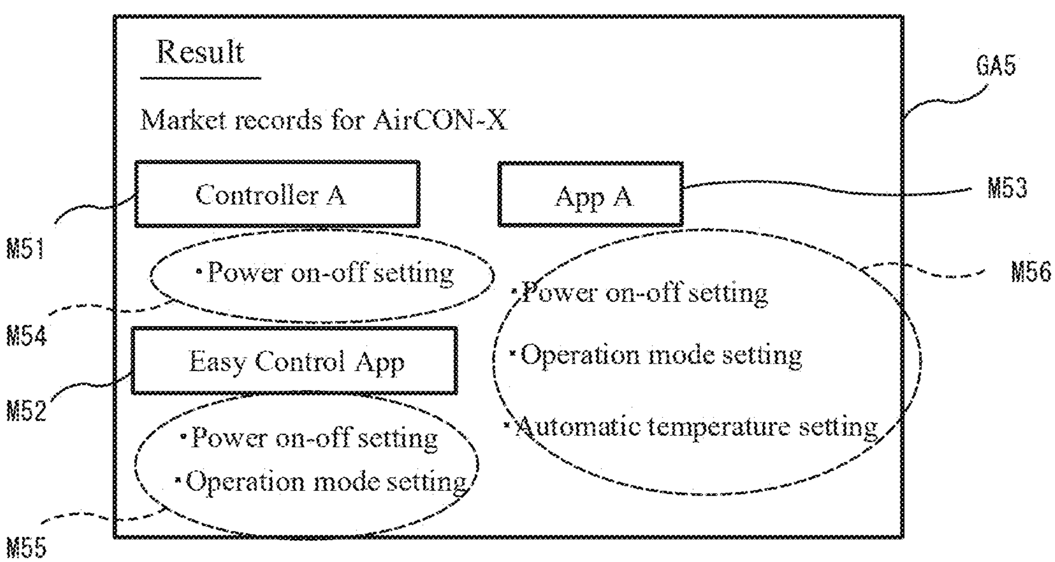

Result
Market records for AirCON-X
Controller A
App A
・Power on-off setting
Easy Control App
・Power on-off setting
・Operation mode setting
・Power on-off setting
・Operation mode setting
・Automatic temperature setting
GA5
M51
M52
M53
M54
M55
M56

FIG.9C

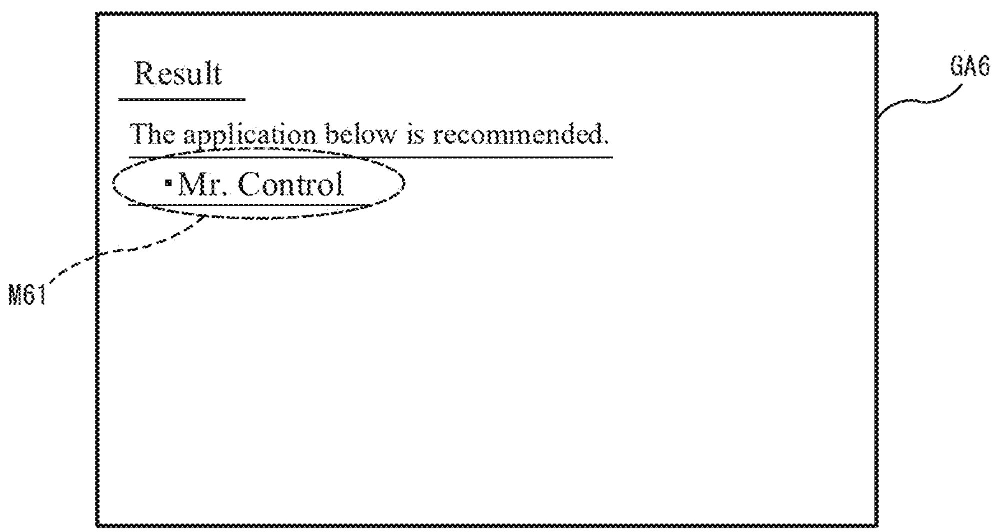

Result
The application below is recommended.
▪Mr. Control
GA6
M61

FIG.10A

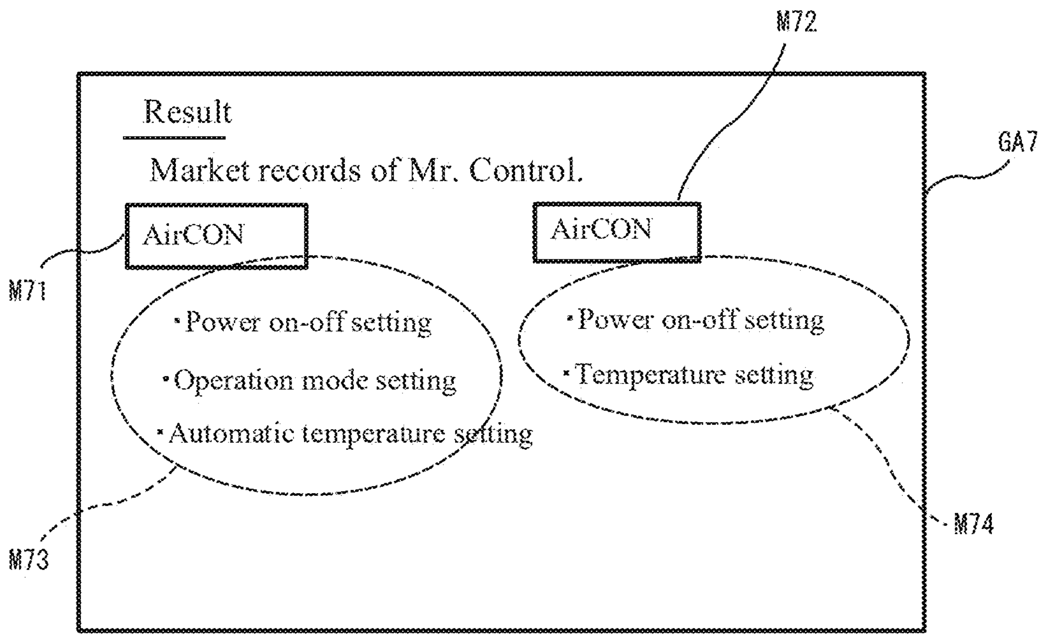

M72
GA7
Result
Market records of Mr. Control.
AirCON
AirCON
M71
・Power on-off setting
・Operation mode setting
・Automatic temperature setting
・Power on-off setting
・Temperature setting
M73
M74

FIG.10B

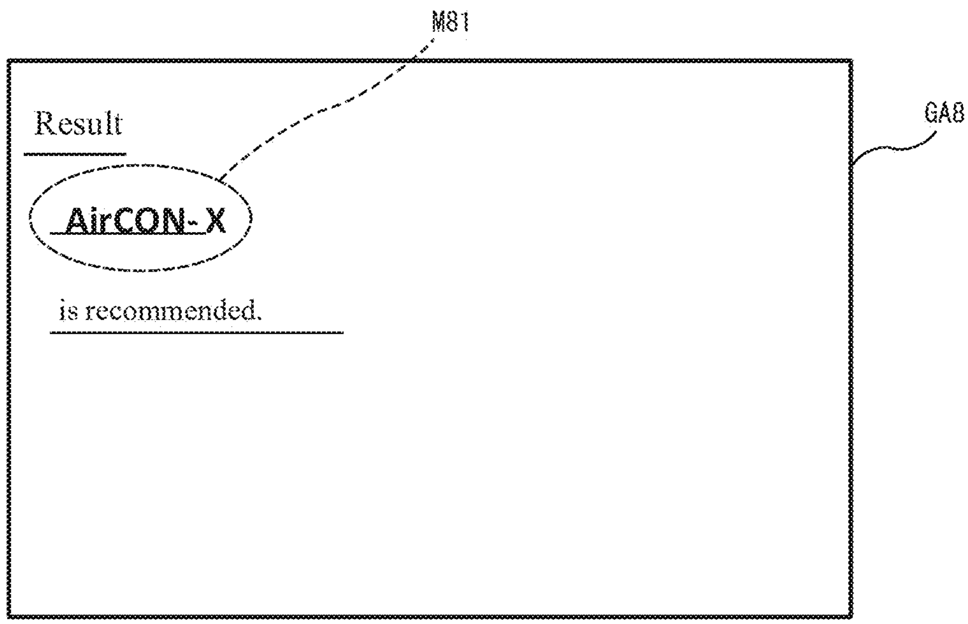


FIG.11

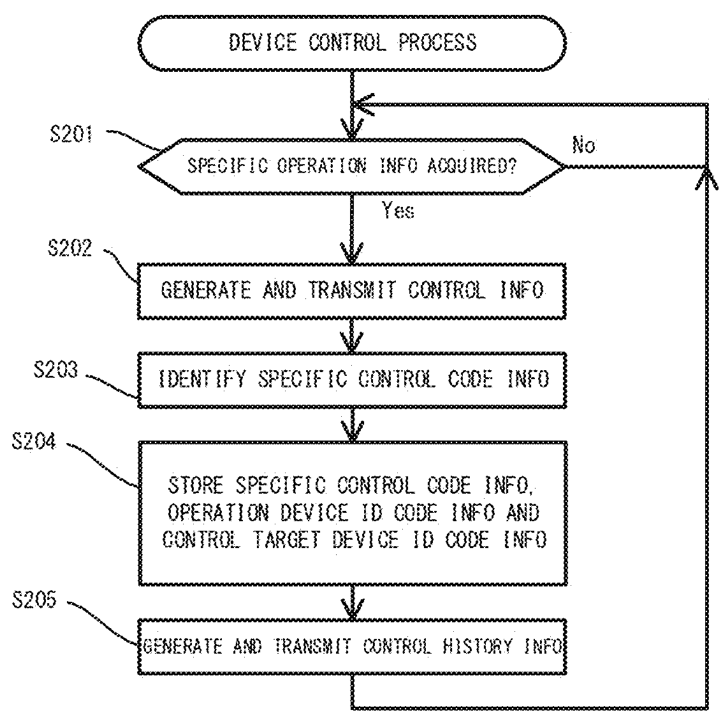

DEVICE CONTROL PROCESS
S201
SPECIFIC OPERATION INFO ACQUIRED?
No
Yes
S202
GENERATE AND TRANSMIT CONTROL INFO
S203
IDENTIFY SPECIFIC CONTROL CODE INFO
S204
STORE SPECIFIC CONTROL CODE INFO, OPERATION DEVICE ID CODE INFO AND CONTROL TARGET DEVICE ID CODE INFO
S205
GENERATE AND TRANSMIT CONTROL HISTORY INFO

FIG.12

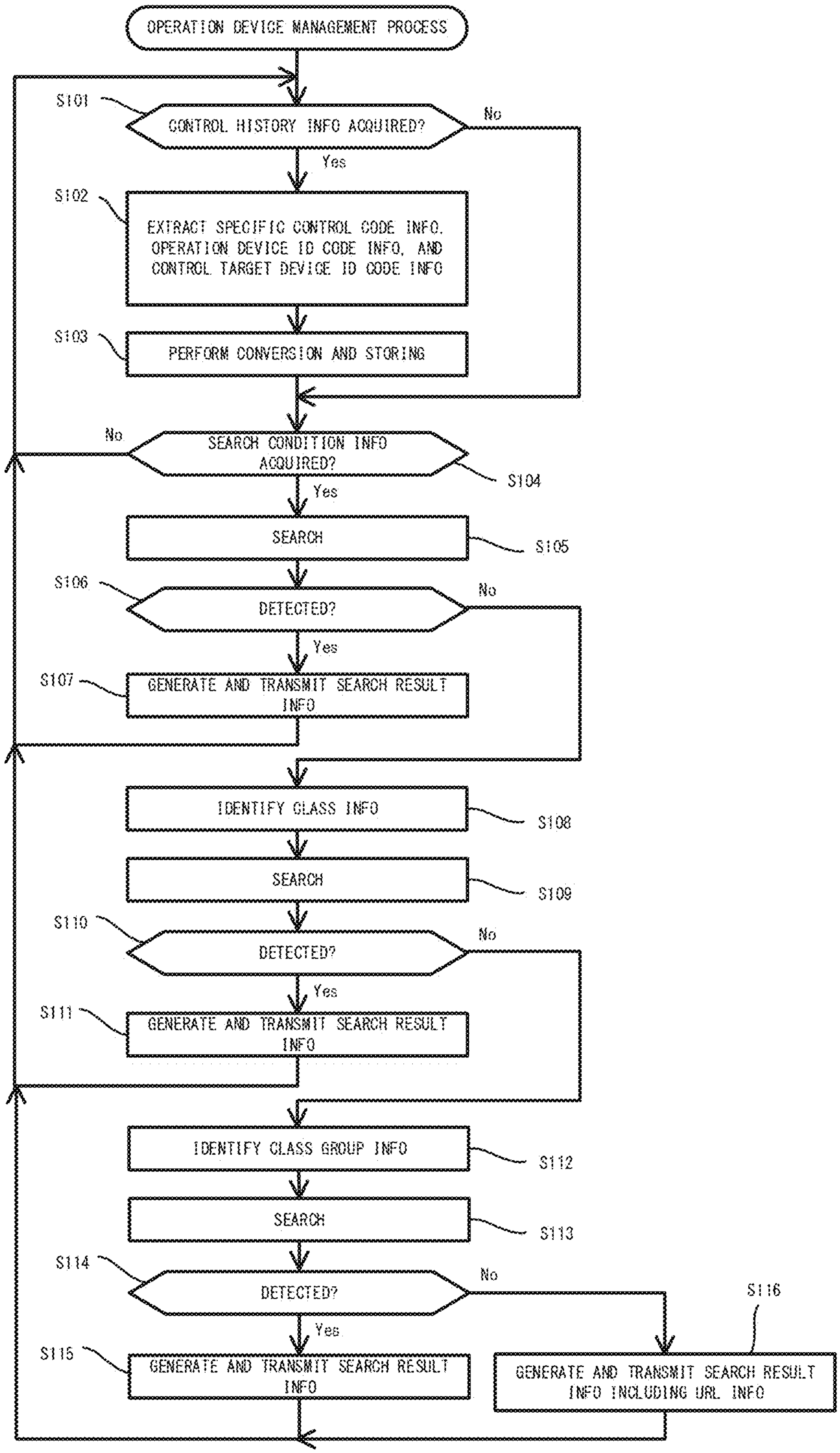

OPERATION DEVICE MANAGEMENT PROCESS
S101
CONTROL HISTORY INFO ACQUIRED?
No
Yes
S102
EXTRACT SPECIFIC CONTROL CODE INFO, OPERATION DEVICE ID CODE INFO, AND CONTROL TARGET DEVICE ID CODE INFO
S103
PERFORM CONVERSION AND STORING
No
SEARCH CONDITION INFO ACQUIRED?
S104
Yes
SEARCH
S105
S106
DETECTED?
No
Yes
S107
GENERATE AND TRANSMIT SEARCH RESULT INFO
IDENTIFY CLASS INFO
S108
SEARCH
S109
S110
DETECTED?
No
Yes
S111
GENERATE AND TRANSMIT SEARCH RESULT INFO
IDENTIFY CLASS GROUP INFO
S112
SEARCH
S113
S114
DETECTED?
No
S116
Yes
S115
GENERATE AND TRANSMIT SEARCH RESULT INFO
GENERATE AND TRANSMIT SEARCH RESULT INFO INCLUDING URL INFO

FIG.13

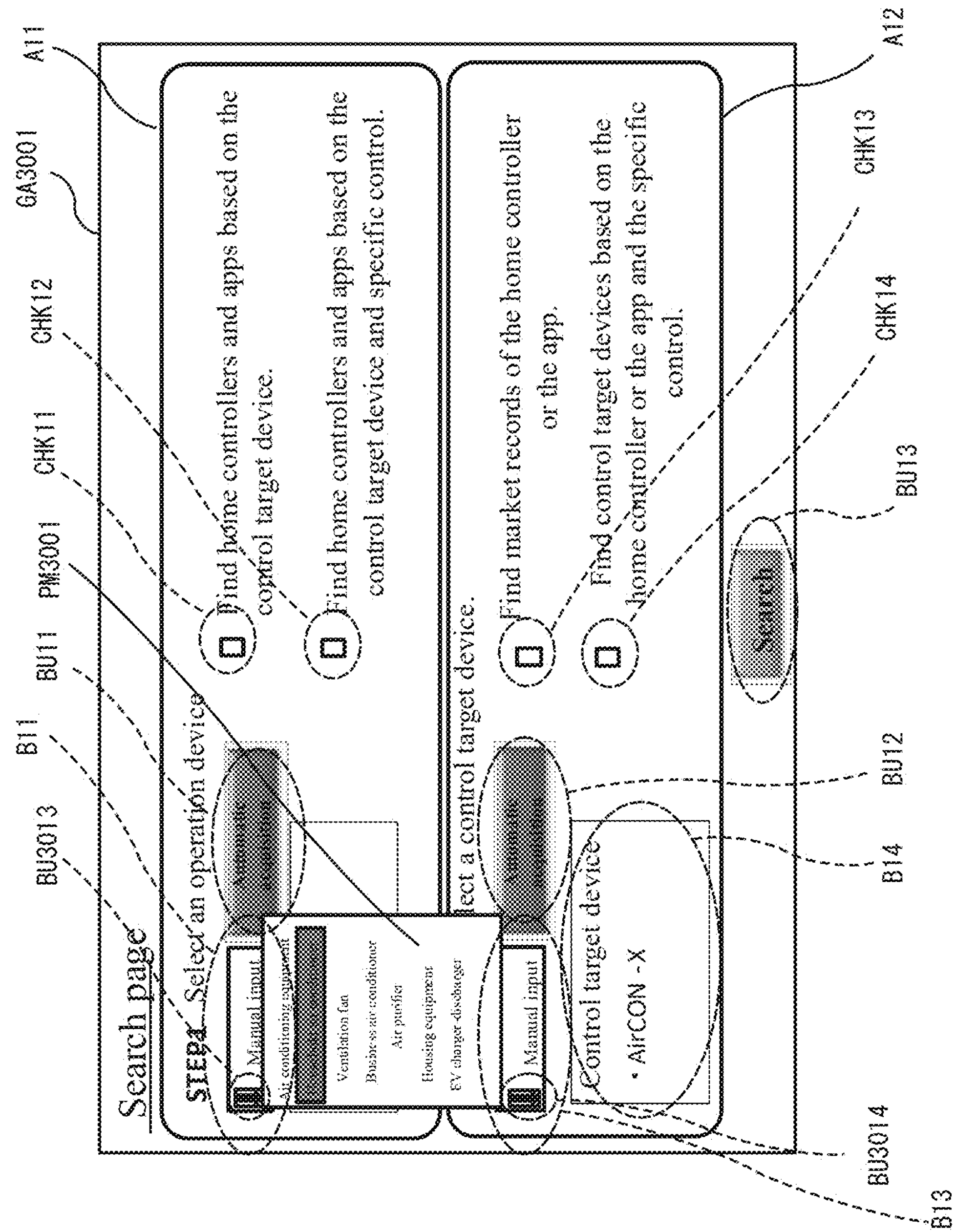

Search page
STEP1 Select an operation device
Manual input
Air conditioning equipment
Ventilation fan
Business air conditioner
Air purifier
Housing equipment
EV charger-discharger
Find home controllers and apps based on the control target device.
Find home controllers and apps based on the control target device and specific control.
ect a control target device.
Manual input
Control target device
• AirCON - X
Find market records of the home controller or the app.
Find control target devices based on the home controller or the app and the specific control.
Search
A11
GA3001
CHK12
CHK11
PM3001
BU11
B11
BU3013
A12
CHK13
CHK14
BU13
BU12
B14
BU3014
B13

FIG.14

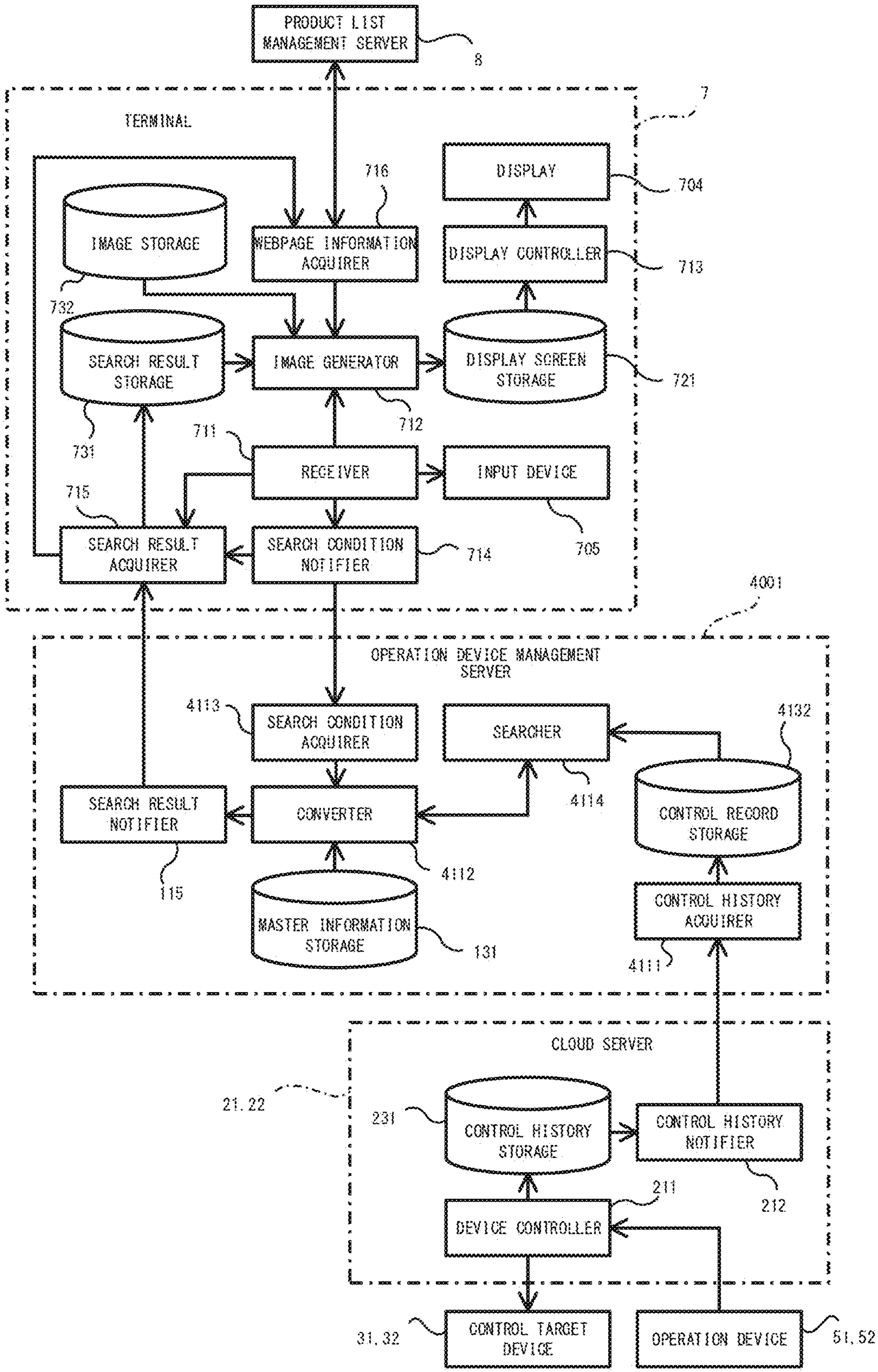
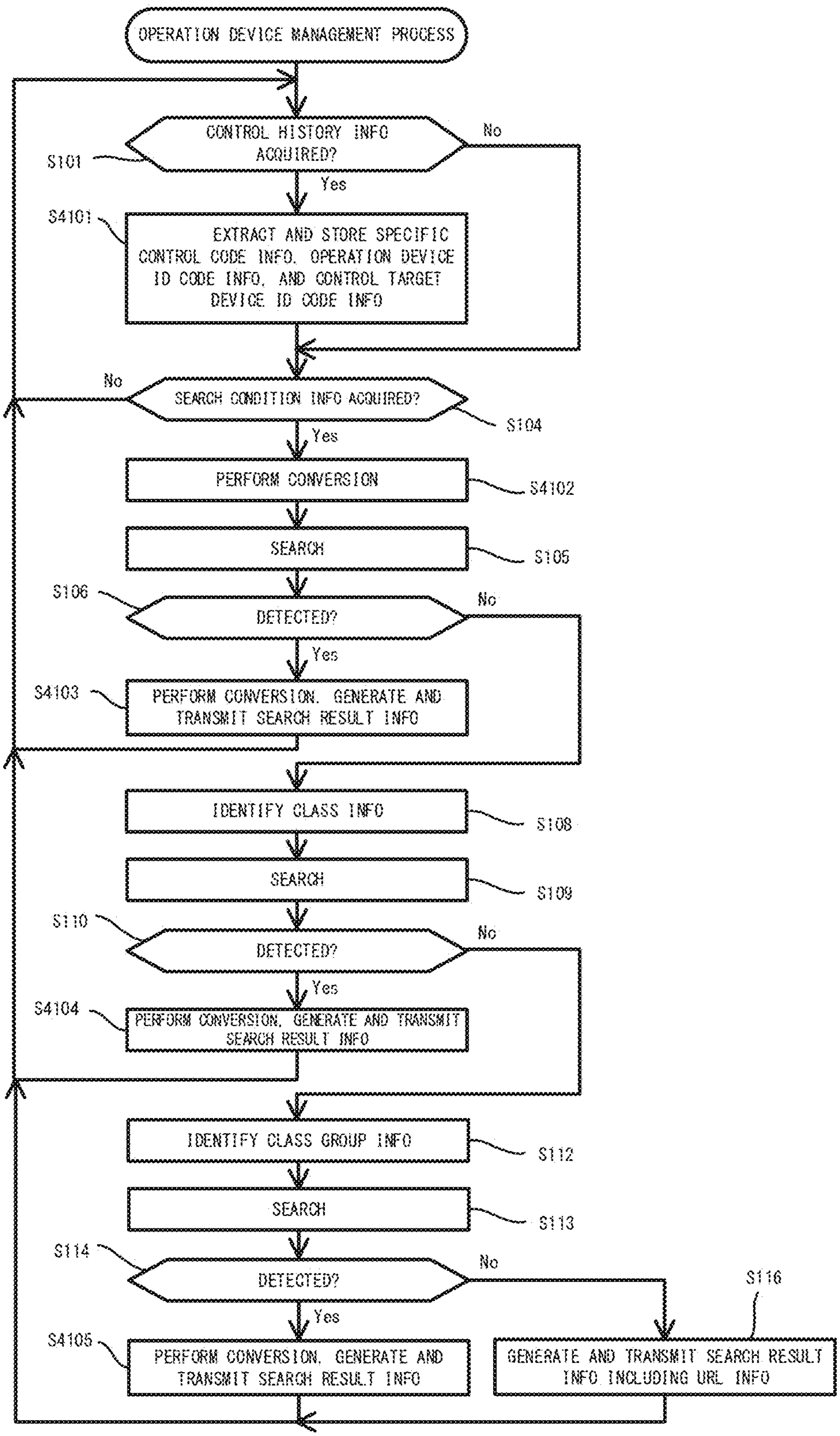

PRODUCT LIST MANAGEMENT SERVER
8
7
TERMINAL
716
DISPLAY
704
IMAGE STORAGE
WEBPAGE INFORMATION ACQUIRER
DISPLAY CONTROLLER
713
732
SEARCH RESULT STORAGE
IMAGE GENERATOR
DISPLAY SCREEN STORAGE
721
731
712
711
RECEIVER
INPUT DEVICE
715
SEARCH RESULT ACQUIRER
SEARCH CONDITION NOTIFIER
714
705
4001
OPERATION DEVICE MANAGEMENT SERVER
4113
SEARCH CONDITION ACQUIRER
SEARCHER
4132
4114
SEARCH RESULT NOTIFIER
CONVERTER
CONTROL RECORD STORAGE
115
4112
CONTROL HISTORY ACQUIRER
MASTER INFORMATION STORAGE
131
4111
CLOUD SERVER
21, 22
231
CONTROL HISTORY STORAGE
CONTROL HISTORY NOTIFIER
211
212
DEVICE CONTROLLER
31, 32
CONTROL TARGET DEVICE
OPERATION DEVICE
51, 52

FIG.15

CONTROL RECORD STORAGE 4132

OPERATION DEVICE ID CODE	CLASS GROUP CODE	CLASS CODE	SPECIFIC CONTROL CODE	CONTROL TARGET DEVICE ID CODE	CLASS GROUP CODE	CLASS CODE
3333	0x05	0xFF	0x80 (OPERATION MODE SETTING)	AABBCCDDEEFFG	0x01	0x30
1111	0x05	0xFF	0x80 (POWER ON-OFF SETTING)	AABBCCDDEEFFG	0x01	0x30
3333	0x05	0xFF	0x80 (POWER ON-OFF SETTING)	AABBCCDDHIJKLM	0x01	0x30
2222	0x05	0xFF	0x81 (AUTOMATIC TEMPERATURE SETTING)	AABBCCDDEEFFG	0x01	0x30
3333	0x05	0xFF	0x80 (POWER ON-OFF SETTING)	AABBCCDDEEFFG	0x01	0x30
3333	0x05	0xFF	0x80 (OPERATION MODE SETTING)	AABBCCDDHIJKLM	0x01	0x30
3333	0x05	0xFF	0x81 (AUTOMATIC TEMPERATURE SETTING)	AABBCCDDEEFFG	0x01	0x30
2222	0x05	0xFF	0x80 (POWER ON-OFF SETTING)	AABBCCDDEEFFG	0x01	0x30
⋮	⋮	⋮	⋮	⋮	⋮	⋮

FIG.16


OPERATION DEVICE MANAGEMENT PROCESS
CONTROL HISTORY INFO ACQUIRED?
S101
No
Yes
S4101
EXTRACT AND STORE SPECIFIC CONTROL CODE INFO, OPERATION DEVICE ID CODE INFO, AND CONTROL TARGET DEVICE ID CODE INFO
No
SEARCH CONDITION INFO ACQUIRED?
S104
Yes
PERFORM CONVERSION
S4102
SEARCH
S105
S106
DETECTED?
No
Yes
S4103
PERFORM CONVERSION, GENERATE AND TRANSMIT SEARCH RESULT INFO
IDENTIFY CLASS INFO
S108
SEARCH
S109
S110
DETECTED?
No
Yes
S4104
PERFORM CONVERSION, GENERATE AND TRANSMIT SEARCH RESULT INFO
IDENTIFY CLASS GROUP INFO
S112
SEARCH
S113
S114
DETECTED?
No
S116
Yes
S4105
PERFORM CONVERSION, GENERATE AND TRANSMIT SEARCH RESULT INFO
GENERATE AND TRANSMIT SEARCH RESULT INFO INCLUDING URL INFO

OPERATION DEVICE MANAGEMENT SYSTEM, OPERATION DEVICE MANAGEMENT APPARATUS, OPERATION DEVICE MANAGEMENT METHOD, AND RECORDING MEDIUM

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a U.S. national stage application of PCT/JP2022/025190 filed on Jun. 23, 2022, the contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to an operation device management system, an operation device management apparatus, an operation device management method, and a non-transitory recording medium storing a program.

BACKGROUND

[0003] An energy management system including a graphical user interface (GUI) apparatus and a device control apparatus that communicates with the GUI apparatus has been developed (see, for example, Patent Literature 1). The device control apparatus includes a collector that collects device information including device manufacturer information from a device connected to a network in a building, a display controller that causes the device name of the device to be displayed on the screen of a user interface device based on the device information collected from the device by the collector, and a registration processor that registers, with a storage, a device with the device name displayed on the screen when the device is selected as a management target.

PATENT LITERATURE

[0004] Patent Literature 1: International Publication No. WO 2016/038882

[0005] For the energy management system described in Patent Literature 1, device control apparatuses have been developed for controlling an operation of a device based on a user operation on the GUI apparatus to control the device operation. Such a system may use an application to allow the GUI apparatus to function as an operation device for operating a device after the device is introduced into the market. An operation device for operating the device may also be introduced to the market subsequently with new functions reflecting the user convenience. A technique is thus awaited for the user to determine an optimum application or an optimum operation device for operating the device owned by the user.

SUMMARY

[0006] Under such circumstances, an objective of the present disclosure is to provide an operation device management system, an operation device management apparatus, an operation device management method, and a program that allow a user to appropriately determine an optimum operation device for operating a control target device owned by the user or an optimum control target device to be controlled through an operation on an operation device owned by the user.

[0007] To achieve the above objective, an operation device management system according to an aspect of the present disclosure includes a terminal and a control device. The control device includes a control record storage to store, in a manner associated with one another, specific control information indicating specific control performed on at least one control target device through an operation performed previously on an operation device, operation device identification information identifying the operation device operated, and control target device identification information identifying the at least one control target device on which the specific control has been performed, and first processing circuitry to acquire search condition information transmitted from the terminal and indicating a search condition for searching for at least one of control target device identification information of a control target device controllable with an operation device operated by a user, operation device identification information of an operation device to control a control target device when operated by a user, or specific control information indicating specific control having a record of being performed on the control target device, to search the control target device identification information, the operation device identification information, and the specific control information stored in the control record storage for at least one of the control target device identification information, the operation device identification information, or the specific control information based on the search condition information, and to generate, when at least one of the control target device identification information, the operation device identification information, or the specific control information is detected, search result information including at least one of the detected control target device identification information, the detected operation device identification information, or the detected specific control information, and transmit the search result information to the terminal. The control record storage further stores, in a manner associated with the control target device identification information, class information indicating a class of the at least one control target device classified based on a type of the at least one control target device. When searching for at least one of the operation device identification information or the control target device identification information based on the search condition and not detecting the operation device identification information or the control target device identification information, the first processing circuitry searches for at least one of operation device identification information of an operation device or control target device identification information of a control target device in a same class as at least one of the operation device being a search target or the control target device being a search target.

[0008] With the technique according to the above aspect of the present disclosure, the searcher searches for, based on the search condition information, at least one of control target device identification information of a control target device controllable with an operation device operated by a user, operation device identification information of an operation device to control a control target device when operated by a user, or specific control information indicating specific control having a record of being performed on the control target device. The search result notifier generates, when at least one of the control target device identification information, the operation device identification information, or the specific control information is detected, search result information including at least one of the detected control target device identification information, the detected operation device identification information, or the detected specific control information, and transmit the search result information to the terminal. With this technique, for the operation device or the control target device owned by the user, a control target device having a record of combined use with the operation device in previous operations performed on the control target device or an operation device having a record of combined use with the control target device in previous operations performed on the control target device can be presented to the user. Thus, the user can appropriately determine an optimum operation device for operating a control target device owned by the user or an optimum control target device to be controlled through an operation on an operation device owned by the user.

BRIEF DESCRIPTION OF DRAWINGS

[0009] FIG. **1** is a schematic diagram of an operation device management system according to an embodiment of the present disclosure, illustrating an example configuration;

[0010] FIG. **2** is a block diagram of the operation device management system according to the embodiment, illustrating the hardware configuration;

[0011] FIG. **3** is a functional block diagram of the operation device management system according to the embodiment;

[0012] FIG. **4** is a diagram illustrating example information stored in a master information storage in an operation device management server according to the embodiment;

[0013] FIG. **5** is a diagram illustrating example information stored in a control record storage in the operation device management server according to the embodiment;

[0014] FIG. **6** is a sequence diagram describing the operation of the operation device management system according to the embodiment;

[0015] FIG. **7** is a diagram of an example operation screen image displayed on a display in a terminal in the embodiment;

[0016] FIG. **8**A is a diagram of an example operation screen image displayed on the display in the terminal in the embodiment;

[0017] FIG. **8**B is a diagram of another example operation screen image displayed on the display in the terminal in the embodiment;

[0018] FIG. **9**A is a diagram of an example operation screen image displayed on the display in the terminal in the embodiment;

[0019] FIG. **9**B is a diagram of another example operation screen image displayed on the display in the terminal in the embodiment;

[0020] FIG. **9**C is a diagram of an example operation screen image displayed on the display in the terminal in the embodiment;

[0021] FIG. **10**A is a diagram of another example operation screen image displayed on the display in the terminal in the embodiment;

[0022] FIG. **10**B is a diagram of still another example operation screen image displayed on the display in the terminal in the embodiment;

[0023] FIG. **11** is a flowchart of an example device control process performed by a cloud server in the embodiment;

[0024] FIG. **12** is a flowchart of an example operation device management process performed by the operation device management server according to the embodiment;

[0025] FIG. **13** is a diagram of another example operation screen image displayed on a display in a terminal in a modification;

[0026] FIG. **14** is a functional block diagram of an operation device management system according to a modification;

[0027] FIG. **15** is a diagram illustrating example information stored in a control record storage in an operation device management server according to the modification; and

[0028] FIG. **16** is a flowchart of an example operation device management process performed by the operation device management server according to the modification.

DETAILED DESCRIPTION

[0029] An operation device management system according to one or more embodiments of the present disclosure is now described with reference to the drawings. The operation device management system includes a terminal, a control record storage, a search condition acquirer, a searcher, and a search result notifier. The control record storage stores, in a manner associated with one another, specific control information indicating specific control performed on at least one control target device through an operation performed previously on an operation device, operation device identification (ID) information identifying the operation device that has been operated, and control target device ID information identifying at least one control target device on which the specific control has been performed. The search condition acquirer acquires search condition information transmitted from the terminal and indicating a search condition for searching for at least one of control target device ID information of a control target device controllable with an operation device operated by a user, operation device ID information of an operation device to control a control target device when operated by the user, or specific control information indicating specific control having a record of being performed on the control target device. The searcher searches for at least one of the control target device ID information, the operation device ID information, or the specific control information based on the search condition information from the control target device ID information, the operation device ID information, and the specific control information stored in the control record storage. The search result notifier generates, when at least one of the control target device ID information, the operation device ID information, or the specific control information is detected, search result information including at least one of the control target device identification information, the operation device identification information, or the specific control information that has been detected, and transmits the search result information to the terminal.

[0030] As illustrated in FIG. **1**, an operation device management system according to the present embodiment includes a terminal **7** connected to one of multiple local area networks NW**21** and NW**22**, and an operation device management server **1** that manages operation devices **51** and **52** connected to the respective local area networks NW**21** and NW**22**. Multiple control target devices **31** and **32** to be controlled using the operation devices **51** and **52** have communication capabilities and are connected to the respective local area networks NW**21** and NW**22**. The local area networks NW**21** and NW**22** are connected to a wide area network NW1 with respective broadband routers (hereafter referred to as BBRs) **41** and **42**. Cloud servers **21** and **22** that control the control target devices **31** and **32** based on operations on the operation devices **51** and **52** by a user are connected to the wide area network NW**1**. A product list management server **8** is also connected to the wide area network NW**1**. The product list management server **8** manages websites showing lists of control target device ID names of the control target devices **31** and **32** and application ID names of operation device applications that are activated on the operation devices **51** and **52**. Each of the local area networks NW**21** and NW**22** is, for example, a wired local area network (LAN) or a wireless LAN. The wide area network NW**1** is, for example, the Internet.

[0031] Each of the control target devices **31** and **32** is, for example, a rice cooker, a ventilation fan, an air conditioner, a water heater, or a bathroom dehumidifier, and operates based on control information transmitted from the cloud server **21** or **22** for controlling the control target device **31** or **32**. Each of the operation devices **51** and **52** is, for example, a smartphone and functions as an operation device for operating the corresponding control target device **31** or **32** with an application being activated for functioning as an operation device. Alternatively, each of the operation devices **51** and **52** is, for example, a remote controller dedicated to operating the corresponding control target device **31** or **32**. Each of the operation devices **51** and **52** generates, when operated by the user, operation information indicating the specific operation and transmits the operation information to the corresponding cloud server **21** or **22**.

[0032] The product list management server **8** manages a website showing a list of control target device ID names of the control target devices **31** and **32** recommended for the operating device applications activated on the operation devices **51** and **52** by manufacturers of the operating device applications, or a website showing a list of operation device ID names identifying the operation device applications recommended for the control target devices **31** and **32** by manufacturers of the control target devices **31** and **32**. The product list management server **8** manages, for example, a website showing a whitelist being a list of operation device applications that guarantee normal operation in an environment in which the operations of the control target devices **31** and **32** are controlled through the cloud servers **21** and **22**, and thus recommended for use by the manufacturers of the control target devices **31** and **32** or the manufacturers of the operation device applications that are activated on the operation devices **51** and **52**. Examples of the whitelist include a list associating the operation device ID names identifying the operation device applications that are activated on the operation devices **51** and **52**, and control target device ID names identifying the control target devices **31** and **32** controllable through an operation on the operation devices **51** and **52** with the operation device applications activated on the operation devices **51** and **52**. When acquiring access request information requesting access to the above websites from the terminal **7**, the product list management server **8** generates webpage information for allowing browse of the websites on the terminal **7** and transmits the webpage information to the terminal **7**.

[0033] As illustrated in FIG. **2**, the terminal **7** is, for example, a personal computer and includes a central processing unit (CPU) **701**, a main storage **702**, an auxiliary storage **703**, a display **704**, an input device **705**, a communicator **706**, and a bus **709** connecting the components to one another. The main storage **702** includes a volatile memory such as a random-access memory (RAM) and is used as a work area for the CPU **701**. The main storage **702** also includes a memory for images (not illustrated) that temporarily stores image information to be displayed on the display **704**. The auxiliary storage **703** is a nonvolatile memory such as a semiconductor flash memory and stores a program for the CPU **701** to perform various processes. The display **704** is a display device such as a liquid crystal display or an organic electroluminescent (EL) display. The input device **705** is, for example, a keyboard. The communicator **706** is connected to the local area networks NW**21** and NW**22** to transmit information transferred from the CPU **701** to the operation device management server **1** through the local area networks NW**21** and NW**22** and the wide area network NW**1**, and to transfer information received from the operation device management server **1** through the wide area network NW**1** and the local area networks NW**21** and NW**22** to the CPU **701**.

[0034] The CPU **701** loads the program stored in the auxiliary storage **703** into the main storage **702** and executes the program to function as a receiver **711**, an image generator **712**, a display controller **713**, a search condition notifier **714**, a search result acquirer **715**, and a webpage information acquirer **716** as illustrated in FIG. **3**. The auxiliary storage **703** illustrated in FIG. **2** includes a search result storage **731** and an image storage **732** as illustrated in FIG. **3**. The memory for images in the main storage **702** illustrated in FIG. **2** includes a display screen storage **721** that temporarily stores screen images to be displayed on the display **704** as illustrated in FIG. **3**. The search result storage **731** stores control target device ID information or operation device ID information included in search result information transmitted from the operation device management server **1**. The control target device ID information identifies the control target device **32** (**31**) having a previous record of being controlled through an operation performed on the operation device **52** (**51**) of the same type as the operation device **51** (**52**) that is controlled by the user. The control target device ID information is input by the user with the input device **705**. The operation device ID information identifies the operation device **52** (**51**) having a previous record of operating the control target device **32** (**31**) of the same type as the control target device **31** (**32**) that is to be operated by the user. The operation device ID information is input by the user with the input device **705**. The image storage **732** stores various items of image information including the image information indicating basic icons for creating an operation screen image.

[0035] The receiver **711** receives a specific operation performed by the user on the input device **705**, generates operation information corresponding to the received specific operation, and provides the operation information to the image generator **712**, the search condition notifier **714**, or the search result acquirer **715**.

[0036] The image generator **712** generates a search screen image for the user to search for the control target device ID information of the control target devices **31** and **32** or the operation device ID information of the operation devices **51** and **52**, or a search result notification screen image for notifying the user of a search result. The image generator **712** generates the search result notification screen image based on the various items of image information stored in the image storage **732** as well as the operation device ID information and the control target device ID information stored in the search result storage **731** in a manner associated with each other. The image generator **712** causes the display screen storage **721** to store the image information indicating the generated search screen image or the search result notification screen image. When the webpage information described below is provided from the webpage information acquirer **716**, the image generator **712** generates a webpage image showing the lists of the control target device ID names and the operation device ID names based on the provided webpage information, and causes the display screen storage **721** to store the image information indicating the generated webpage image. The display controller **713** causes the display **704** to display the search screen image or the search result notification screen image described above based on the image information stored in the display screen storage **721**.

[0037] With the display **704** displaying the search screen image, the search condition notifier **714** generates the search condition information indicating search conditions for searching for the control target device ID information or the operation device ID information based on information input by the user with the input device **705**. More specifically, the search condition notifier **714** generates the search condition information including the operation device ID information of the operation device **51** or **52**, or the control target device ID information of the control target device **31** or **32** input by the user. The search condition notifier **714** then transmits the generated search condition information to the operation device management server **1**. The search condition notifier **714** also provides the operation device ID information or the control target device ID information input by the user to the search result acquirer **715**.

[0038] When acquiring the search result information transmitted from the operation device management server **1**, the search result acquirer **715** extracts the control target device ID information or the operation device ID information included in the acquired search result information, and causes the search result storage **731** to store the extracted control target device ID information or operation device ID information. When acquiring the search result information including the control target device ID information from the operation device management server **1**, the search result acquirer **715** causes the search result storage **731** to store the control target device ID information extracted from the acquired search result information in a manner associated with the operation device ID information provided from the search condition notifier **714**. When acquiring the search result information including the operation device ID information from the operation device management server **1**, the search result acquirer **715** causes the search result storage **731** to store the operation device ID information extracted from the acquired search result information in a manner associated with the control target device ID information provided from the search condition notifier **714**. When the search result information includes uniform resource locator (URL) information of a website managed by the product list management server **8**, the search result acquirer **715** extracts the URL information from the search result information and provides the URL information to the webpage information acquirer **716**.

[0039] When the URL information is provided from the search result acquirer **715**, the webpage information acquirer **716** transmits the access request information described above to the product list management server **8** based on the provided URL information. The webpage information acquirer **716** then acquires the webpage information transmitted from the product list management server **8** and provides the acquired webpage information to the image generator **712**.

[0040] As illustrated in FIG. **2**, the cloud servers **21** and **22** each include a CPU **201**, a main storage **202**, an auxiliary storage **203**, a communicator **206**, and a bus **209** connecting the components to one another. The CPU **201** is, for example, a multicore processor. The main storage **202** includes a volatile memory such as a RAM and is used as a work area for the CPU **201**. The auxiliary storage **203** is a large-capacity nonvolatile memory and stores a program for implementing various functions of the cloud server **21** or **22**. The communicator **206** is connected to the wide area network NW**1**.

[0041] The CPU **201** loads the program stored in the auxiliary storage **203** into the main storage **202** and executes the program to function as a device controller **211** and a control history notifier **212** as illustrated in FIG. **3**. The auxiliary storage **203** illustrated in FIG. **2** includes a control history storage **231** as illustrated in FIG. **3**. The control history storage **231** stores specific control code information indicating specific control performed on the control target device **31** or **32** based on the operation information indicating a specific operation transmitted from the operation device **51** or **52** when the user performs an operation on the operation device **51** or **52**. The control history storage **231** stores the specific control code information in a manner associated with operation device ID code information of the operation device **51** or **52**, or operation device ID code information of the operation device application activated on the operation device **51** or **52**, and with control target device ID code information of the control target device **31** or **32**. The specific control code information is, for example, expressed in a first format using EPC parameters defined in the ECHONET Lite (registered trademark) standard. More specifically, for the specific control being power on-off setting, for example, the specific control information is set to 0x80. The operation device ID information is, for example, expressed in the first format using the operation device ID code assigned to the operation device application that is activated on the operation device **51** or **52**. The control target device ID information is expressed in the first format using the control target device ID code assigned to the control target device **31** or **32**.

[0042] When acquiring, from the operation device **51** or **52**, specific operation information indicating a specific operation performed by the user on the operation device **51** or **52**, the device controller **211** controls the control target device **31** or **32** based on the acquired specific operation information. The device controller **211** controls the operation of the control target device **31** or **32** by generating the control information for controlling the control target device **31** or **32** based on the specific operation information and transmitting the control information to the control target device **31** or **32**. The control information includes the specific control code information described above. Each time the device controller **211** generates and transmits the control information to the control target device **31** or **32**, the device controller **211** identifies the specific control code information included in the control information and causes the control history storage **231** to store the identified specific control code information in a manner associated with the operation device ID code information of the operation device **51** or **52** that is the source of the specific operation information and the control target device ID code information of the control target device **31** or **32** that is the destination of the control information.

[0043] The control history notifier **212** generates control history information including the specific control code information, the operation device ID code information, or the operation device ID code information and the control target device ID code information stored in the control history storage **231**, and transmits the generated control history information to the operation device management server **1**. The control history notifier **212** generates and transmits the control history information to the operation device management server **1** each time the control history storage **231** newly stores the specific control code information, the operation device ID code information, or the operation device ID code information and the control target device ID code information in a manner associated with one another. Alternatively, for example, the control history notifier **212** may generate, at each predetermined control history notification time, the control history information including the specific control code information, the operation device ID code information, or the operation device ID code information and the control target device ID code information newly stored in the control history storage **231** during a period from the immediately preceding control history notification time and the present control history notification time, and transmit the control history information to the operation device management server **1**.

[0044] As illustrated in FIG. **2**, the operation device management server **1** includes a CPU **101**, a main storage **102**, an auxiliary storage **103**, a communicator **106**, and a bus **109** connecting the components to one another. The CPU **101** is, for example, a multicore processor. The main storage **102** includes a volatile memory such as a RAM and is used as a work area for the CPU **101**. The auxiliary storage **103** is a large-capacity nonvolatile memory and stores a program for implementing various functions of the operation device management server **1**. The communicator **106** is connected to the wide area network NW1.

[0045] The CPU **101** loads the program stored in the auxiliary storage **103** into the main storage **102** and executes the program to function as a control history acquirer **111**, a converter **112**, a search condition acquirer **113**, a searcher **114**, and a search result notifier **115** as illustrated in FIG. **3**. The auxiliary storage **103** illustrated in FIG. **2** includes a master information storage **131** and a control record storage **132** as illustrated in FIG. **3**. The master information storage **131** stores information indicating the correspondence between the sets of information in the first format described above and sets of information in a second format different from the first format. As illustrated in FIG. **4**, for example, the master information storage **131** stores manufacturer name information indicating the name of the manufacturer of the control target device **31** or **32** or the name of the manufacturer of the operation device application activated on the operation device **51** or **52** in a manner associated with manufacturer code information indicating a manufacturer code preassigned to the manufacturer. The master information storage **131** stores control target device ID name information identifying the control target device **31** or **32** in the second format in a manner associated with the control target device ID code information indicating the control target device ID code in the first format assigned to the control target device **31** or **32**. The master information storage **131** stores the control target device ID name information and the control target device ID code information in a manner associated with class code information indicating a class of the control target device **31** or **32** as classified based on the type and class group code information. The master information storage **131** stores, in a manner associated with the operation device ID code information indicating the operation device ID code in the first format assigned to the operation device **51** or **52** or the operation device application, operation device ID name information indicating the operation device ID name identifying, in the second format, the above operation device **51** or **52**, or the operation device application that is activated on the operation device **51** or **52**. The master information storage **131** stores the operation device ID name information and the operation device ID code information in a manner associated with the class code information indicating the class of the operation device as classified based on the type and the class group code information. The master information storage **131** also stores the specific control information indicating the specific control performed on the control target device **31** or **32** in a manner associated with the specific control code information assigned to different types of specific control.

[0046] Referring back to FIG. **3**, the control record storage **132** stores, in a manner associated with one another, the specific control information indicating the specific control performed on the corresponding control target device **31** or **32** through an operation performed previously on the operation device **51** or **52** by the user, the operation device ID name information identifying the operation device **51** or **52** that has been operated by the user, and the control target device ID name information identifying the corresponding control target device **31** or **32**. As illustrated in FIG. **5**, for example, the control record storage **132** stores the specific control information indicating the specific control performed on the control target device **31** or **32** through an operation performed previously on the operation device **51** or **52**, together with the operation device ID name information of the corresponding operation device **51** or **52** and the control target device ID name information of the control target device **31** or **32** in a manner associated with the class code information and the class group code information of the operation device **51** or **52** and the control target device **31** or **32**. Although not illustrated in FIG. **5**, the control record storage **132** also stores the manufacturer name information indicating the manufacturer name of the operation device **51** or **52** and of the control target device **31** or **32** in a manner associated with the corresponding operation device ID name information and the control target device ID name information.

[0047] Referring back to FIG. **3**, the control history acquirer **111** acquires control history information including the specific control code information indicating the specific control performed on the control target device **31** or **32**, the operation device ID code information identifying the operation device **51** or **52** operated by the user, and the control target device ID code information identifying the control target device **31** or **32**. The control history acquirer **111** extracts the specific control code information, the operation device ID code information, and the control target device ID code information included in the acquired control history information and provides the extracted sets of information to the converter **112**.

[0048] The converter **112** converts the specific control code information, the operation device ID code information, and the control target device ID code information provided from the control history acquirer **111** respectively to the specific control information, the operation device ID name information, and the control target device ID name information, and causes the control record storage **132** to store the resultant sets of information. The converter **112** identifies, among the specific control information, the operation device ID name information, and the control target device ID name information stored in the master information storage **131**, the specific control information, the operation device ID name information, and the control device ID name information associated with the specific control code information, the operation device ID code information, and the control target device ID code information provided from the control history acquirer **111**. The converter **112** then causes the control record storage **132** to store the identified specific control information, the operation device ID name information, and the control target device ID name information. The converter **112** also refers to the manufacturer name information stored in the master information storage **131** to identify the manufacturer name information associated with the manufacturer code information provided from the control history acquirer **111**, and causes the control record storage **132** to store the identified manufacturer name information in a manner associated with the operation device ID name information or the control target device ID name information.

[0049] The search condition acquirer **113** acquires the search condition information transmitted from the terminal **7** indicating the search conditions for searching for the control target device ID name information of the control target device **31** or **32** that is controllable by the operation device **51** or **52** operated by the user. Alternatively, the search condition acquirer **113** acquires the search condition information indicating the search conditions for searching for the operation device ID name information of the operation device **51** or **52** that can control the control target device **31** or **32** when operated by the user. The search condition acquirer **113** then provides the acquired search condition information to the searcher **114**.

[0050] The searcher **114** searches for, based on the search condition information provided from the search condition acquirer **113**, at least one of the control target device ID name information of the control target device **32** (**31**) having a record of being controlled through an operation performed on the operation device **52** (**51**) of the same type as the operation device **51** (**52**) that is specified in the search condition information, or the specific control information indicating the specific control in the record. Alternatively, the searcher **114** searches for at least one of the operation device ID name information of the operation device **52** (**51**) having a record of being operated to control the control target device **32** (**31**) of the same type as the control target device **31** (**32**) that is specified in the search condition information, or the specific operation information indicating the specific operation in the record. Alternatively, the searcher **114** searches for the specific control information indicating the specific control performed with the operation device **52** (**51**) of the same type as the operation device **51** (**52**) specified in the above search condition information on the control target device **32** (**31**) of the same type as the control target device **31** (**32**) specified in the above search condition information. In other words, when the search condition information includes the operation device ID name information and the control target device ID name information to be used as search keys, the searcher **114** searches for the specific control information using the operation device ID name information and the control target device ID name information as search keys. When the search condition information simply includes the control target device ID name information to be used as a search key, the searcher **114** searches for the operation device ID name information and the specific control information using the control target device ID name information as a search key. When the search condition information includes the control target device ID name information and the specific control information to be used as search keys, the searcher **114** searches for the operation device ID name information using the control target device ID name information and the specific control information as search keys. When the search condition information simply includes the operation device ID name information to be used as a search key, the searcher **114** searches for the control target device ID name information and the specific control information using the operation device ID name information as a search key. When the search condition information includes the operation device ID name information and the specific control information to be used as search keys, the searcher **114** searches for the control target device ID name information using the operation device ID name information and the specific control information as search keys.

[0051] When searching for the control target device ID name information of the control target device **32** (**31**) having a record of being controlled through an operation performed on the operation device **52** (**51**) of the same type as the operation device **51** (**52**) that is operated by the user and detecting no control target device ID name information, the searcher **114** searches for the control target device ID name information of the control target device **32** (**31**) with the same class code information assigned as the class code information of the control target device **32** (**31**). When further searching for the control target device ID name information of the control target device **32** (**31**) with the same class code information assigned as the class code information of the control target device **32** (**31**) and detecting no control target device ID name information, the searcher **114** searches for the control target device ID name information of the control target device **32** (**31**) with the same class group code information assigned as the class group code information of the control target device **32** (**31**). When detecting at least one of the control target device ID name information, the operation device ID name information, or the specific control information through the search, the searcher **114** provides at least one of the control target device ID name information, the operation device ID name information, or the specific control information that has been detected to the search result notifier **115**.

[0052] When at least one of the control target device ID name information, the operation device ID name information, or the specific control information is provided from the searcher **114**, the search result notifier **115** generates the search result information including the provided control target device ID name information or the operation device ID name information and transmits the search result information to the terminal **7**. When the control target device ID name information or the operation device ID name information is not detected by the searcher **114**, the search result notifier **115** generates the search result information including the URL information indicating the URL of a website managed by the product list management server **8** and transmits the search result information to the terminal **7**. The URL information indicates, for example, the URL of a website showing a list of control target device ID names of the control target device **31** or **32** recommended for the operation device **51** or **52** by the manufacturer of the operation device **51** or **52**, or the URL of a website showing a list of operation device ID names recommended for operating the control target device **31** or **32** by the manufacturer of the control target device **31** or **32**.

[0053] The operation of the operation device management system according to the present embodiment is now described with reference to FIGS. **6** to **10**B. As illustrated in FIG. **6**, when the user first performs an operation to control the operation of the control target device **31** or **32** on the operation device **51** or **52**, the cloud server **21** or **22** acquires the operation information transmitted from the operation device **51** or **52** (step S**1**). The cloud server **21** or **22** generates the control information for controlling the control target device **31** or **32** based on the acquired specific operation information (step S**2**). The generated control information is transmitted from the cloud server **21** or **22** to the control target device **31** or **32** (step S**3**). When acquiring the control information, the control target device **31** or **32** starts operating based on the acquired control information (step S**4**). The cloud server **21** or **22** identifies the specific control code information indicating the specific control performed on the control target device **31** or **32** (step S**5**). The cloud server **21** or **22** then causes the control history storage **231** to store the generated specific control information in a manner associated with the operation device ID code information of the operation device **51** or **52** that is the source of the specific operation information or the operation device application activated on the operation device **51** or **52**, and with the control target device ID code information of the control target device **31** or **32** that is the destination of the control information (Step S**6**). The cloud server **21** or **22** then generates the control history information including the specific control code information, the operation device ID code information, and the control target device ID code information stored in the control history storage **231** (step S**7**). The generated control history information is transmitted from the cloud server **21** or **22** to the operation device management server **1** (step S**8**).

[0054] When acquiring the control history information, the operation device management server **1** extracts the specific control code information, the operation device ID code information, and the control target device ID code information included in the acquired control history information (step S**9**). The operation device management server **1** then refers to the information stored in the master information storage **131** to convert the extracted specific control code information, operation device ID code information, and control target device ID code information respectively to the specific control information, the operation device ID name information, and the control target device ID name information, and causes the control record storage **132** to store the resultant sets of information (step S**10**).

[0055] In an example, the user also performs a search preparation operation on the input device **705** in the terminal **7** to search for the control target device ID name information of the control target device **31** or **32**, or the operation device ID name information of the operation device **51** or **52**. In this case, the terminal **7** generates the search screen image for the user to input the search conditions and causes the display **704** to display the search screen image (step S**11**).

[0056] The terminal **7** causes the display **704** to display, for example, a search screen image GA**1** illustrated in FIG. **7**. The search screen image GA**1** includes an area A**11**, an area A**12**, and a button image BU**13**. The search screen image GA**1** specifies the operation device ID name information used as a key in searching for the specific control performed on the control target device **31** or **32**, or the control target device ID name information of the control target device **31** or **32**. The area A**12** specifies the control target device ID name information used as a key in searching for the specific control performed on the control target device **31** or **32**, or the operation device ID name information of the operation device **51** or **52**. The button image BU**13** displays Search to be selected to perform a search. The area A**11** includes a display field B**12** for displaying the operation device ID name used as a search key, an input field B**11** for inputting the operation device name used as a search key, and a button image BU**11** to be selected to automatically acquire the operation device ID name of the operation device **51** or **52** connected to the local area network NW**21** or NW**22** to which the terminal **7** is connected. The area A**12** includes a display field B**14** for displaying the control target device name used as a search key, an input field B**13** for inputting the control target device ID name used as a search key, and a button image BU**12** to be selected to automatically acquire the control target device name of the control target device **31** or **32** connected to the local area network NW**21** or NW**22** to which the terminal **7** is connected.

[0057] The area A**11** further includes checkbox images CHK**11** and CHK**12** to be selected in searching without using the operation device name as a search key. When the checkbox image CHK**11** is selected, the operation device ID name and the specific control are searched for using the control target device ID name as a search key. When the checkbox image CHK**12** is selected, the operation device ID name is searched for using the control target device ID name and the specific control as search keys. The area A**12** also includes checkbox images CHK**13** and CHK**14** to be selected in searching without using the control target device name as a search key. When the checkbox image CHK**13** is selected, the control target device ID name and the specific control are searched for using the operation device ID name as a search key. When the checkbox image CHK**14** is selected, the control target device ID name is searched for using the operation device ID name and the specific control as search keys.

[0058] When the user selects the checkbox image CHK**12** and then the button image BU**13** with the input device **705**, with the search screen image GA**1** displayed on the display **704**, a search screen image GA**2** illustrated in FIG. **8**A is displayed on the display **704**. The search screen image GA**2** includes checkbox images CHK**21**, CHK**22**, CHK**23**, CHK**24**, CHK**25**, and CHK**26**, and a button image BU**21** displaying Search to be selected to perform a search. The checkbox images CHK**21**, CHK**22**, CHK**23**, CHK**24**, CHK**25**, and CHK**26** are used as search keys together with the control target device name to respectively select the power on-off setting, temperature setting, automatic temperature setting, operation mode setting, cooling mode setting, and heating mode setting as the specific control to be performed on the control target device **31** or **32**. In the example illustrated in FIG. **8**A, the power on-off setting, the automatic temperature setting, and the operation mode setting are selected as search keys.

[0059] When the user selects the checkbox image CHK**14** and then the button image BU**13** with the input device **705**, with the search screen image GA**1** illustrated in FIG. **7** displayed on the display **704**, a search screen image GA**3** illustrated in FIG. **8**B is displayed on the display **704**. The search screen image GA**3** includes checkbox images CHK**31**, CHK**32**, CHK**33**, CHK**34**, CHK**35**, and CHK**36**, and a button image BU**31** displaying Search to be selected to perform a search. The checkbox images CHK**31**, CHK**32**, CHK**33**, CHK**34**, CHK**35**, and CHK**36** are used as search keys together with the operation device ID name to respectively select the power on-off setting, the temperature setting, the automatic temperature setting, the operation mode setting, the cooling mode setting, and the heating mode setting as the specific control to be performed on the control target device **31** or **32**. In the example illustrated in FIG. **8**B, the power on-off setting, the automatic temperature setting, and the operation mode setting are used as search keys.

[0060] As illustrated in FIG. **6**, when the user performs the above search key specifying operation and search start operation with the search screen image displayed on the display **704** in the terminal **7**, the terminal **7** generates the above search condition information (step S**12**). In the search key specifying operation, as described above, the user specifies, with the input device **705**, the operation device ID name, the control target device ID name, and the specific control used as search keys with the search screen image displayed on the display **704** in the terminal **7**. In the search start operation, for example, the user selects, with the input device **705**, the button image BU**13** without selecting the checkbox image CHK**11**, CHK**12**, CHK**13**, or CHK**14** or after selecting the checkbox images CHK**11** and CHK **13** with the search screen image GA**1** illustrated in FIG. **7** displayed on the display **704**. In the search start operation, for example, the user selects, with the input device **705**, at least one of the checkbox image CHK**21**, CHK**22**, CHK**23**, CHK**24**, CHK**25**, or CHK**26** and then the button image BU**21** with the search screen image GA**2** illustrated in FIG. **8**A displayed on the display **704**. Alternatively, the user selects, with the input device **705**, at least one of the checkbox image CHK**31**, CHK**32**, CHK**33**, CHK**34**, CHK**35**, or CHK**36** and then the button image BU**31** with the search screen image GA**3** illustrated in FIG. **8**B displayed on the display **704**.

[0061] In an example, the user selects, with the input device **705**, the operation device ID name and the control target device ID name used as search keys without selecting any of the checkbox image CHK**11**, CHK**12**, CHK**13** or CHK**14** and then selects the button image BU**13** with the search screen image GA**1** illustrated in FIG. **7** displayed on the display **704** in the terminal **7**. In this case, the terminal **7** generates the search condition information including the operation device ID name information and the control target device ID name information used as search keys. In an example, the user selects, with the input device **705**, the control target device ID name used as a search key and the checkbox image CHK**11** and then selects the button image BU**13** with the search screen image GA**1** illustrated in FIG. **7** displayed on the display **704** in the terminal **7**. In this case, the terminal **7** generates the search condition information simply including the control target device ID name information used as a search key. In an example, the user selects, with the input device **705**, the operation device ID name used as a search key and the checkbox image CHK**13** and then selects the button image BU**13** with the search screen image GA**1** illustrated in FIG. **7** displayed on the display **704** in the terminal **7**. In this case, the terminal **7** generates the search condition information simply including the operation device ID name information used as a search key. In an example, the user selects, with the input device **705**, the specific control used as a search key and then the button image BU**21** with the search screen image GA**2** illustrated in FIG. **8**A displayed on the display **704** in the terminal **7**. In this case, the terminal **7** generates the search condition information including the control target device ID name information and the specific control information used as search keys. In an example, the user selects, with the input device **705**, the specific control used as a search key and then the button image BU**31** with the search screen image GA**3** illustrated in FIG. **8**B displayed on the display **704** in the terminal **7**. In this case, the terminal **7** generates the search condition information including the operation device ID name information and the specific control information used as search keys. Referring back to FIG. **6**, the generated search condition information is then transmitted from the terminal **7** to the operation device management server **1** (step S**13**).

[0062] When acquiring the search condition information, the operation device management server **1** searches for at least one of the control target device ID name information, the operation device ID name information, or the specific control information based on the acquired search condition information (step S**14**). The operation device management server **1** searches for the control target device ID name information of the control target device **32** (**31**) having a record of being controlled through an operation performed on the operation device **52** (**51**) of the same type as the operation device **51** (**52**) that is operated by the user based on the search condition information. Alternatively, the operation device management server **1** searches for the operation device ID name information of the operation device **52** (**51**) having a record of being operated to control the control target device **32** (**31**) of the same type as the control target device **31** (**32**) that is to be controlled by the user based on the search condition information. When searching for the control target device ID name information of the control target device **32** (**31**) having a record of being controlled through an operation performed on the operation device **52** (**51**) of the same type as the operation device **51** (**52**) that is operated by the user based on the search condition information and detecting no control target device ID name information, the operation device management server **1** searches for the control target device ID name information of the control target device **32** (**31**) with the same class code information assigned as the class code information of the control target device **32** (**31**). When searching for the control target device ID name information of the control target device **32** (**31**) with the same class code information assigned as the class code information of the control target device **32** (**31**) and detecting no control target device ID name information, the operation device management server **1** searches for the control target device ID name information of the control target device **32** (**31**) with the same class group code information assigned as the class group code information of the control target device **32** (**31**).

[0063] When detecting at least one of the control target device ID name information, the operation device ID name information, or the specific control information through the search, the operation device management server **1** generates the search result information including at least one of the control target device ID name information, the operation device ID name information, or the specific control information that has been detected (step S**15**). When detecting none of the control target device ID name information, the operation device ID name information, or the specific control information through the search, the operation device management server **1** generates the search result information including the URL information described above. The generated search result information is then transmitted from the operation device management server **1** to the terminal **7** (step S**16**).

[0064] When acquiring the search result information, the terminal **7** extracts at least one of the control target device ID information, the operation device ID information, or the specific control information included in the acquired search result information, generates a search result display screen image based on at least one of the control target device ID information, the operation device ID information, or the specific control information that has been extracted, and causes the display **704** to display the search result display screen image on the display **704** (step S**17**).

[0065] When the search condition information transmitted to the operation device management server **1** includes the operation device ID name information and the control target device ID name information used as search keys, the terminal **7** acquires the search result information including the specific control information detected through the search using the operation device ID name information and the control target device ID name information as search keys. The terminal **7** then generates, for example, a search result display screen image GA**4** illustrated in FIG. **9**A including a message M**41** indicating the detected specific control and causes the display **704** to display the search result display screen image GA**4**. When the search condition information transmitted to the operation device management server **1** simply includes the control target device ID name information used as a search key, the terminal **7** acquires the search result information including the operation device ID name information and the specific control information detected through the search using the control target device ID name information as a search key. The terminal **7** then generates, for example, a search result display screen image GA**5** illustrated in FIG. **9**B including, in a manner associated with one another, a message M**51** indicating the detected operation device ID name information of the detected operation device **51** or **52** and messages M**52** and M**53** each indicating the operation device ID name information, and messages M**54**, M**55**, and M**56** each indicating the specific control information indicating the specific control performed with the operation device **51** or **52**, or the operation device application, and causes the display **704** to display the search result display screen image GA**5**. When the search condition information transmitted from the operation device management server **1** includes the control target device ID name information and the specific control information used as search keys, the terminal **7** acquires the search result information including the operation device ID name information detected through the search using the control target device ID name information and the specific control information as search keys. The terminal **7** then generates, for example, a search result display screen image GA**6** illustrated in FIG. **9**C including a message M**61** indicating the detected operation device ID name information and causes the display **704** to display the search result display screen image GA**6**.

[0066] When the search condition information transmitted from the operation device management server **1** simply includes the operation device ID name information used as a search key, the terminal **7** acquires the search result information including the control target device ID name information and the specific control information detected through the search using the operation device ID name information as a search key. The terminal **7** then generates, for example, a search result display screen image GA**7** illustrated in FIG. **10**A including, in a manner associated with one another, messages M**71** and M**72** each indicating the detected control target device ID name information and messages M**73** and M**74** each indicating the specific control information performed on each control target device **31** or **32** and causes the display **704** to display the search result display screen image GA**7**. When the search condition information transmitted from the operation device management server **1** includes the operation device ID name information and the specific control information used as search keys, the terminal **7** acquires the search result information including the control target device ID name information detected through the search using the operation device ID name information and the specific control information as search keys. The terminal **7** then generates, for example, a search result display screen image GA**8** illustrated in FIG. **10**B including a message M**81** indicating the detected control target device ID name information and causes the display **704** to display the search result display screen image GA**8**.

[0067] When the acquired search result information includes the URL information of a website managed by the product list management server **8**, the terminal **7** extracts the URL information from the search result information and acquires the webpage information from the product list management server **8** based on the extracted URL information. The terminal **7** then generates, based on the acquired webpage information, a webpage image showing a list of control target device ID names recommended by the manufacturer of the operation device application or a list of operation device ID names recommended by the manufacturer of the control target device **31** or **32**, and causes the display **704** to display the webpage image.

[0068] A device control process performed by the cloud server **21** or **22** in the present embodiment for controlling the control target device **31** or **32** based on the specific operation information acquired from the operation device **51** or **52** is now described with reference to FIG. **11**. The device control process starts in response to the activation of a program for performing the device control process in the cloud server **21** or **22**. The device controller **211** first determines whether the specific operation information transmitted from the operation device **51** or **52** is acquired (step S**201**). Each time the device controller **211** determines that no specific operation information is acquired (No in step S**201**), the device controller **211** repeats the processing in step S**201**. When determining that the specific operation information is acquired (Yes in step S**201**), the device controller **211**

generates the control information for controlling the control target device **31** or **32** based on the acquired specific operation information and transmits the control information to the control target device **31** or **32** (step S**202**). The device controller **211** then identifies the specific control code information corresponding to the generated control information (step S**203**). Subsequently, the device controller **211** causes the control history storage **231** to store, in a manner associated with one another, the identified specific control code information, the operation device ID code information of the operation device **51** or **52** that is the source of the specific operation information or the operation device application activated on the operation device **51** or **52**, or the operation device ID code information of the operation device **51** or **52**, and the control target device ID code information of the control target device **31** or **32** that is the destination of the control information. The control history notifier **212** then generates the control history information including the specific control code information, the operation device ID code information, and the control target device ID code information stored in the control history storage **231** and transmits the generated control history information to the operation device management server **1** (step S**205**). The processing in step S**201** is then performed again.

[0069] An operation device management process performed by the operation device management server **1** according to the present embodiment is now described with reference to FIG. **12**. The operation device management process starts in response to the activation of a program for performing the operation device management process in the operation device management server **1**. The control history acquirer **111** first determines whether the control history information transmitted from the cloud server **21** or **22** is acquired (step S**101**). When the control history acquirer **111** determines that no control history information is acquired (No in step S**101**), the processing in step S**104** (described later) is performed. When determining that the control history information is acquired (Yes in step S**101**), the control history acquirer **111** extracts the specific control code information, the operation device ID code information, and the control target device ID code information included in the acquired control history information (step S**102**). The converter **112** then refers to the information stored in the master information storage **131** to convert the extracted specific control code information, operation device ID code information, and control target device ID code information respectively to the specific control information, the operation device ID name information, and the control target device ID name information, and causes the control record storage **132** to store the resultant sets of information (step S**103**).

[0070] Subsequently, the search condition acquirer **113** determines whether the search condition information transmitted from the terminal **7** is acquired (step S**104**). When the search condition acquirer **113** determines that no search condition information is acquired (No in step S**104**), the processing in step S**101** is performed again. In an example, the search condition acquirer **113** determines that the search condition information is acquired (Yes in step S**104**). In this case, the searcher **114** searches for at least one of the control target device ID name information, the operation device ID name information, and the specific control information based on the acquired search condition information (step S**105**). The searcher **114** performs the search using the operation device ID name information, the control target device ID name information, or the specific control information included in the search condition information as a search key. The searcher **114** then determines whether at least one of the operation device ID name information, the control target device ID name information, or the specific control information is detected through the search (step S**106**). In an example, the searcher **114** determines that at least one of the operation device ID name information, the control target device ID name information, or the specific control information is detected (Yes in step S**106**). In this case, the search result notifier **115** generates the search result information including at least one of the control target device ID name information, the operation device ID name information, or the specific control information that has been detected, and transmits the search result information to the terminal **7** (step S**107**). The processing in step S**101** is then performed again.

[0071] In an example, the searcher **114** determines that none of the operation device ID name information, the control target device ID name information, or the specific control information is detected (No in step S**106**). In this case, the searcher **114** refers to the class information stored in the master information storage **131** to identify the class information corresponding to the operation device ID name information or the control target device ID name information used as a search key (step S**108**). The searcher **114** then searches the various information items stored in the control record storage **132** for the operation device ID name information and the control target device ID name information corresponding to the identified class information (step S**109**). Subsequently, the searcher **114** determines whether at least one of the operation device ID name information, the control target device ID name information, or the specific control information is detected through the search (step S**110**). In an example, the searcher **114** determines that at least one of the operation device ID name information, the control target device ID name information, or the specific control information is detected (Yes in step S**110**). In this case, the search result notifier **115** generates the search result information including at least one of the control target device ID name information, the operation device ID name information, or the specific control information that has been detected, and transmits the search result information to the terminal **7** (step S**111**). The processing in step S**101** is then performed again.

[0072] In an example, the searcher **114** determines that none of the operation device ID name information, the control target device ID name information, or the specific control information is detected (No in step S**110**). In this case, the searcher **114** refers to the class group information stored in the master information storage **131** to identify the class group information corresponding to the operation device ID name information or the control target device ID name information used as a search key (step S**112**). The searcher **114** then searches the various information items stored in the control record storage **132** for the operation device ID name information or the control target device ID name information corresponding to the identified class group information (step S**113**). The searcher **114** then determines whether at least one of the operation device ID name information, the control target device ID name information, or the specific control information is detected through the search (step S**114**). In an example, the searcher **114** determines that at least one of the operation device ID name information, the control target device ID name information, or the specific control information is detected (Yes in step S**114**). In this case, the search result notifier **115** generates the search result information including at least one of the control target device ID name information, the operation device ID name information, or the specific control information that has been detected, and transmits the search result information to the terminal **7** (step S**115**). The processing in step S**101** is then performed again.

[0073] In an example, the searcher **114** determines that none of the operation device ID name information, the control target device ID name information, or the specific control information is detected (No in step S**114**). In this case, the search result notifier **115** generates the search result information including the URL information indicating the URL of a website managed by the product list management server **8** and transmits the search result information to the terminal **7** (step S**116**). The processing in step S**101** is then performed again.

[0074] In the operation device management system according to the present embodiment, as described above, the searcher **114** searches for, based on the search condition information, at least one of the control target device ID name information of the control target device **31** or **32** controllable with the operation device **51** or **52** operated by the user, the operation device ID name information of the operation device **51** or **52** that can control the control target device when operated by a user, or the specific control information indicating the specific control having a record of being performed on the control target device. When at least one of the control target device ID name information, the operation device ID name information, or the specific control information is detected, the search result notifier **115** generates the search result information including at least one of the control target device ID name information, the operation device ID name information, or the specific control information that has been detected and transmits the search result information to the terminal **7**. With this system, for the operation device **51** or **52** owned by the user, the control target device **31** or **32** having a record of combined use with the operation device **51** or **52** in previous operations performed on the control target device **31** or **32** can be presented to the user, and for the control target device **31** or **32** owned by the user, the operation device **51** or **52** having a record of combined use with the control target device **31** or **32** in previous operations performed on the control target device **31** or **32** can be presented to the user. The user can thus appropriately determine the optimum operation device **51** or **52** for operating the control target device **31** or **32** owned by the user or the optimum control target device **31** or **32** to be controlled through an operation performed on the operation device **51** or **52** owned by the user.

[0075] Techniques have recently been developed for controlling the control target device **31** or **32** with the cloud server **21** or **22** through the wide area network NW**1** as described above. Techniques have also been developed for controlling the control target device **31** or **32** with a smartphone on which the operation device application is activated, or for operating the control target device **31** or **32** with a control device connected to the local area network NW**21** or NW**22** to which the control target device **31** or **32** is connected. The manufacturers of the multiple control target devices **31** and **32** or the operation devices **51** and **52** have been provided the control target devices **31** and **32** or the operation devices **51** and **52** based on standardized communication standards such as ECHONET Lite (registered trademark). Thus, the operation device **51** or **52** with a manufacturer different from the manufacturer of the control target device **31** or **32** has been increasingly used as the operation device **51** or **52** for operating the control target device **31** or **32**. Additionally, version updates may add new functions to or remove old functions from the operation device applications that are activated on the operation devices **51** and **52**. A technique has thus been awaited for the user to easily determine the compatibility between the operation device **51** or **52** and the control target device **31** or **32**.

[0076] The operation device management system according to the present embodiment can present, to the user, the operation device **51** or **52** having a record of combined use with the control target device **31** or **32** in previous operations performed on the control target device **31** or **32**, or the control target device **31** or **32** having a record of combined use with the operation device **51** or **52** in previous operations performed on the control target device **31** or **32**. The user can thus appropriately determine a combination of the operation device **51** or **52** and the control target device **31** or **32** with such a previous record as having high compatibility.

[0077] Additionally, the control record storage **132** in the present embodiment stores the class information indicating the class of the control target device **31** or **32** as classified based on the type of the control target device **31** or **32** in a manner associated with the control target device ID name information of the corresponding control target device **31** or **32**. When searching for at least one of the operation device ID name information or the control target device ID name information based on the search condition and not detecting the operation device ID name information or the control target device ID name information, the searcher **114** searches for at least one of the operation device ID name information of the operation device or the control target device ID name information of the control target device in the same class as at least one of the operation device being a search target or the control target device being a search target. This can eliminate resetting the operation device ID name information or the control target device ID name information used as a search key by the user, thus reducing the operation load on the user.

[0078] Additionally, when the control target device ID name information or the operation device ID name information is not detected, the search result notifier **115** generates the search result information including the URL information indicating the URL of a website managed by the product list management server **8** and transmits the search result information to the terminal **7**. This can eliminate searching for a website showing an associated product list by the user when the control target device ID name information or the operation device ID name information is not detected.

[0079] Although the embodiment of the present disclosure has been described above, the present disclosure is not limited to the above embodiment. For example, the user may select, as a search key, the class information indicating the class of the operation device **51** or **52** or the control target device **31** or **32**, or the class group information indicating the class group of the operation device **51** or **52** or the control target device **31** or **32**. The searcher **114** may then search for at least one of the specific control information, the operation device ID name information, or the control target device ID name information using at least one of all operation devices **51** and **52**, or all control target devices **31** and **32** in the selected class as search keys. For example, when the class including the operation device **51** or **52** is selected, the searcher **114** searches for at least one of the specific control information or the control target device ID name information using all operation devices **51** and **52** in the selected class as search keys.

[0080] In the present modification, the terminal **7** may cause the display **704** to display, for example, a search screen image GA**3001** illustrated in FIG. **13**. In FIG. **13**, like reference signs denote like components in FIG. **7** in the embodiment. The search screen image GA**3001** includes a button image BU**3013** to be selected to display a pop-up menu image PM**3001** located adjacent to the left end of the input field B**11** for inputting the operation device name used as a search key, and a button image BU**3014** to be selected to display a pop-up menu image PM**3001** located adjacent to the left end of the input field B**13** for inputting the control target device ID name used as a search key. When the user selects, with the input device **705**, the button image BU**3013** or BU**3014** with the search screen image GA**3001** displayed on the display **704** in the terminal **7**, the pop-up menu image PM**3001** is displayed. In the example illustrated in FIG. **13**, the pop-up menu image PM**3001** includes pre-registered class names of Household air conditioner, Ventilation fan, Business air conditioner, Air purifier, and EV charger-discharger each indicating a class of the control target device **31** or **32** and pre-registered class group names of Air conditioning equipment and Housing equipment each indicating a class group.

[0081] This structure allows a search using the class or the class group of the operation device **51** or **52**, or the control target device **31** or **32**, thus improving the search efficiency for the user.

[0082] In the embodiment, the converter **112** converts the specific control code information, the operation device ID code information, and the control target device ID code information respectively to the specific control information, the operation device ID name information, and the control target device ID name information, and causes the control record storage **132** to store the resultant sets of information. The searcher **114** then searches for at least one of the control target device ID name information, the operation device ID name information, or the specific control information based on the search condition information. In some embodiments, however, the searcher may search for at least one of the specific control code information, the operation device ID code information, or the control target device ID code information based on the search condition information, and the converter may convert at least one of the specific control code information, the operation device ID code information, or the control target device ID code information that has been detected to the specific control information, the operation device ID name information, or the control target device ID name information.

[0083] As illustrated in FIG. **14**, an operation device management server **4001** according to the present modification has the same hardware configuration as the operation device management server **1** described in the embodiment, and functions as a control history acquirer **4111**, a converter **4112**, a search condition acquirer **4113**, a searcher **4114**, and a search result notifier **115**. In FIG. **14**, like reference signs denote like components in FIG. **3** in the embodiment. The operation device management server **4001** includes a master information storage **131** and a control record storage **4132**. As illustrated in FIG. **15**, for example, the control record storage **4132** stores the specific control code information indicating the specific control performed on the control target device **31** or **32** through an operation performed previously on the operation device **51** or **52**, together with the operation device ID code information of the corresponding operation device **51** or **52** and the control target device ID code information of the control target device **31** or **32**, in a manner associated with the class code information and the class group code information of the operation device **51** or **52** and the control target device **31** or **32**.

[0084] Referring back to FIG. **14**, the control history acquirer **4111** acquires the control history information transmitted from the cloud server **21** or **22**, extracts the specific control code information, the operation device ID code information, and the control target device ID code information included in the acquired control history information, and causes the control record storage **4132** to store the extracted sets of information in a manner associated with one another.

[0085] When acquiring the search condition information transmitted from the terminal **7**, the search condition acquirer **4113** provides the acquired search condition information to the converter **4112**.

[0086] When acquiring the search condition information from the search condition acquirer **4113**, the converter **4112** refers to the information stored in the master information storage **131** to convert at least one of the operation device ID name information, the control target device ID name information, or the specific control information used as a search key included in the search condition information to the operation device ID code information, the control target device ID code information, or the specific control code information. The converter **4112** then provides the operation device ID code information, the control target device ID code information, or the specific control code information to the searcher **4114**. When at least one of the operation device ID code information, the control target device ID code information, or the specific control code information that has been detected by the searcher **4114** is provided from the searcher **4114**, the converter **4112** refers to the information stored in the master information storage **131** to convert at least one of the operation device ID code information, the control target device ID code information, or the specific control code information that has been provided to the operation device ID name information, the control target device ID name information, or the specific control information. The converter **4112** then provides the operation device ID name information, the control target device ID name information, or the specific control information to the search result notifier **115**.

[0087] The searcher **4114** searches for the control target device ID code information of the control target device **32** (**31**) having a record of being controlled through an operation performed on the operation device **52** (**51**) of the same type as the operation device **51** (**52**) that is operated by the user based on at least one of the operation device ID code information, the control target device ID code information, or the specific control code information used as a search key provided from the converter **4112**. Alternatively, the searcher **4114** searches for the operation device ID code information of the operation device **52** (**51**) or the operation device application having a record of being operated to control the control target device **32** (**31**) of the same type as the control target device **31** (**32**) that is to be controlled by the user. Alternatively, the searcher **4114** searches for the specific control code information indicating the specific control performed on the control target device **32** (**31**) of the same type as the control target device **31** (**32**) intended to be controlled by the user with the operation device **52** (**51**) of the same type as the operation device **51** (**52**) operated by the user. When detecting at least one of the control target device ID code information, the operation device ID code information, or the specific control code information through the search, the searcher **4114** provides at least one of the control target device ID code information, the operation device ID code information, or the specific control code information that has been detected to the converter **4112**.

[0088] When at least one of the control target device ID name information, the operation device ID name information, or the specific control information is provided from the converter **4112**, the search result notifier **115** generates the search result information including the provided control target device ID name information or the operation device ID name information and transmits the search result information to the terminal **7**.

[0089] An operation device management process performed by the operation device management server **4001** according to the present modification is now described with reference to FIG. **16**. In the operation device management process illustrated in FIG. **16**, like reference signs denote like processing in FIG. **12** in the embodiment. The control history acquirer **4111** first determines whether the control history information transmitted from the cloud server **21** or **22** is acquired (step S**101**). When the control history acquirer **4111** determines that no control history information is acquired (No in step S**101**), the processing in step S**104** (described later) is performed. When determining that the control history information is acquired (Yes in step S**101**), the control history acquirer **4111** extracts the specific control code information, the operation device ID code information, and the control target device ID code information included in the acquired control history information and causes the control record storage **4132** to store the extracted sets of information (step S**4101**).

[0090] The search condition acquirer **4113** then determines whether the search condition information transmitted from the terminal **7** is acquired (step S**104**). When the search condition acquirer **4113** determines that no search condition information is acquired (No in step S**104**), the processing in step S**101** is performed again. In an example, the search condition acquirer **4113** determines that the search condition information is acquired (Yes in step S**104**). In this case, the converter **4112** refers to the information stored in the master information storage **131** to convert at least one of the operation device ID name information, the control target device ID name information, or the specific control information used as a search key included in the acquired search condition information to the operation device ID code information, the control target device ID code information, or the specific control code information (step S**4102**). Subsequently, the searcher **4114** searches for at least one of the control target device ID code information, the operation device ID code information, or the specific control code information based on at least one of the operation device ID code information, the control target device ID code information, or the specific control code information used as a search key provided from the converter **4112** (step S**105**). The searcher **4114** then determines whether at least one of the operation device ID code information, the control target device ID code information, or the specific control code information is detected through the search (step S**106**). In an example, the searcher **4114** determines that at least one of the operation device ID code information, the control target device ID code information, or the specific control code information is detected (Yes in step S**106**). In this case, the converter **4112** refers to the information stored in the master information storage **131** to convert at least one of the operation device ID code information, the control target device ID code information, or the specific control code information that has been detected to the operation device ID name information, the control target device ID name information, or the specific control information. The search result notifier **115** generates the search result information including at least one of the device ID name information, the operation device ID name information, or the specific control information that has been converted and transmits the search result information to the terminal **7** (step S**4103**). The processing in step S**101** is then performed again.

[0091] In an example, the searcher **4114** determines that none of at least one of the operation device ID code information, the control target device ID code information, or the specific control code information is detected (No in step S**106**). In this case, a series of processing from step S**108** to step S**110** is performed. In an example, the searcher **4114** then determines, in step S**110**, that at least one of the operation device ID code information, the control target device ID code information, or the specific control code information is detected (Yes in step S**110**). In this case, the converter **4112** refers to the information stored in the master information storage **131** to convert at least one of the operation device ID code information, the control target device ID code information, or the specific control code information that has been detected to the operation device ID name information, the control target device ID name information, or the specific control information. The search result notifier **115** generates the search result information including at least one of the control target device ID name information, the operation device ID name information, or the specific control information that has been converted and transmits the search result information to the terminal **7** (step S**4104**). The processing in step S**101** is then performed again.

[0092] In an example, the searcher **4114** determines that none of the operation device ID name information, the control target device ID name information, or the specific control information is detected (No in step S**110**). In this case, a series of processing from step S**112** to step S**114** is performed. In an example, the searcher **4114** then determines, in step S**114**, that at least one of the operation device ID code information, the control target device ID code information, or the specific control code information is detected (Yes in step S**114**). In this case, the converter **4112** refers to the information stored in the master information storage **131** to convert at least one of the operation device ID code information, the control target device ID code information, or the specific control code information that has been detected to the operation device ID name information, the control target device ID name information, or the specific control information. In this case, the search result notifier **115** generates the search result information including at least one of the device ID name information, the operation device ID name information, or the specific control information that has been detected, and transmits the search result information to the terminal **7** (step S**4105**). The processing in step S**101** is then performed again.

[0093] When the searcher **4114** determines that none of the operation device ID name information, the control target device ID name information, or the specific control information is detected (No in step S**114**), the processing in step S**116** is performed.

[0094] With this structure, the operation device ID code information, the control target device ID code information, or the specific control code information detected by the searcher **4114** alone is converted to the operation device ID name information, the control target device ID name information, or the specific control information. This can reduce the number of information items to be converted among the operation device ID code information, the control target device ID code information, and specific control code information, thus reducing the conversion load on the converter **4112**.

[0095] In some embodiments, the terminal **7** may include an ID information acquirer that acquires the operation device ID name information of all operation devices **51** and **52** connected to the local area networks NW**21** and NW**22** to which the terminal **7** is connected, or the operation device ID name information of all operation device applications activated on the operation devices **51** and **52**, and the control target device ID name information of all control target devices **31** and **32**. When the ID information acquirer acquires the operation device ID name information of all operation devices **51** and **52** or the operation device ID name information of all operation device applications activated on the operation devices **51** and **52** and the control target device ID name information, the search condition notifier **714** may generate, based on the acquired sets of information, the search condition information using the acquired sets of information as search keys and transmit the search condition information to the operation device management server **1**. In some embodiments, the ID information acquirer may acquire the operation device ID code information of the operation device **51** or **52** connected to the local area network NW**21** or NW**22** to which the terminal **7** is connected, and the control target device ID code information of the control target device **31** or **32**.

[0096] This structure can eliminate inputting the operation device ID name, the application ID name, or the control target device ID name used as a search key, thus improving the search efficiency for the user.

[0097] The functions of the operation device management server **1** or **4001**, the cloud servers **21** and **22**, and the terminal **7** according to one or more embodiments of the present disclosure may be implemented with a common computer system, rather than with a dedicated system. For example, a computer connected to a network may store a program for performing the above operations in a non-transitory computer-readable recording medium (for example, a compact disc read-only memory, or CD-ROM) for distribution. The program may be installed in the computer system to implement the operation device management server **1** or **4001**, the cloud servers **21** and **22**, and the terminal **7** that perform the above processing.

[0098] The program may be provided to the computer in any manner. For example, the program may be uploaded to a bulletin board system (BBS) through a communication line and distributed to a computer. The computer then activates the program and executes the program in the same manner as other applications under the control of the operating system (OS). The computer thus functions as the operation device management server **1** or **4001**, the cloud servers **21** and **22**, and the terminal **7**.

[0099] Although the embodiment of the present disclosure and the modifications of the embodiment have been described above, the present disclosure is not limited to the embodiment and the modifications. The present disclosure includes the embodiment and the modifications combined as appropriate and changed as appropriate.

INDUSTRIAL APPLICABILITY

[0100] One or more embodiments of the present disclosure are applicable to an operation device management system that presents the compatibility between an operation device and a control target device to be controlled through an operation on the operation device.

1. An operation device management system, comprising:
a terminal; and
a control device, wherein
the control device includes
- a control record storage to store, in a manner associated with one another, specific control information indicating specific control performed on at least one control target device through an operation performed previously on an operation device, operation device identification information identifying the operation device operated, and control target device identification information identifying the at least one control target device on which the specific control has been performed, and
- first processing circuitry to
  - acquire search condition information transmitted from the terminal, the search condition information indicating a search condition for searching for at least one of control target device identification information of a control target device controllable with an operation device operated by a user, operation device identification information of an operation device to control a control target device when operated by a user, or specific control information indicating specific control having a record of being performed on the control target device,
  - search the control target device identification information, the operation device identification information, and the specific control information stored in the control record storage for at least one of the control target device identification information, the operation device identification information, or the specific control information based on the search condition information, and
  - generate, when at least one of the control target device identification information, the operation device identification information, or the specific control information is detected, search result information including at least one of the detected control target device identification information, the detected operation device identification infor-

mation, or the detected specific control information, and transmit the search result information to the terminal,

the control record storage further stores, in a manner associated with the control target device identification information, class information indicating a class of the at least one control target device classified based on a type of the at least one control target device, and

when searching for at least one of the operation device identification information or the control target device identification information based on the search condition and not detecting the operation device identification information or the control target device identification information, the first processing circuitry searches for at least one of operation device identification information of an operation device or control target device identification information of a control target device in a same class as at least one of the operation device being a search target or the control target device being a search target.

2. The operation device management system according to claim **1**, wherein

the search condition information includes at least one of the operation device identification information, the control target device identification information, or the specific control information used as a search key.

3. The operation device management system according to claim **2**, wherein

the control record storage further stores, in a manner associated with the control target device identification information, class information indicating a class of the at least one control target device classified based on a type of the at least one control target device.

4. The operation device management system according to claim **2**, wherein

when searching for at least one of the operation device identification information or the control target device identification information based on the search condition and not detecting the operation device identification information or the control target device identification information, the first processing circuitry searches for at least one of operation device identification information of an operation device or control target device identification information of a control target device in a same class as at least one of the operation device being a search target or the control target device being a search target.

5. The operation device management system according to claim **1**, wherein

the first processing circuitry transmits, when the control target device identification information or the operation device identification information is detected, uniform resource locator information indicating a predetermined uniform resource locator to the terminal.

6. The operation device management system according to claim **1**, wherein

the first processing circuitry

controls, when acquiring, from the operation device, specific operation information indicating a specific operation performed on the operation device by the user, the at least one control target device based on the acquired specific operation information,

acquires control history information including specific control information expressing the specific control performed on the at least one control target device in a first format, operation device identification information identifying the operation device operated by the user and expressed in the first format, and control target device identification information identifying the at least one control target device and expressed in the first format, and

converts the specific control information, the operation device identification information, and the control target device identification information included in the control history information respectively to specific control information, operation device identification information, and control target device identification information expressed in a second format different from the first format, and causes the control record storage to store the specific control information, the operation device identification information, and the control target device identification information expressed in the second format.

7. The operation device management system according to claim **3**, wherein

the first processing circuitry

controls, when acquiring, from the operation device, specific operation information indicating a specific operation performed on the operation device by the user, the at least one control target device based on the acquired specific operation information,

acquires control history information including specific control information expressing the specific control performed on the at least one control target device in a first format, operation device identification information identifying the operation device operated by the user and expressed in the first format, and control target device identification information identifying the at least one control target device and expressed in the first format, and

converts, when the searcher detects the control target device identification information or the operation device identification information, the detected control target device identification information or the detected operation device identification information to control target device identification information or operation device identification information expressed in a second format different from the first format,

the control record storage stores the specific control information, the operation device identification information, and the control target device identification information in the first format in a manner associated with one another,

the first processing circuitry searches for the control target device identification information or the operation device identification information expressed in the first format based on the search condition information including the operation device identification information and the specific control information expressed in the first format or including the control target device identification information and the specific control information expressed in the first format, and

the first processing circuitry transmits, to the terminal, the operation device identification information or the control target device identification information resulting from conversion and expressed in the second format.

8. The operation device management system according to claim **1**, wherein
the terminal includes
second processing circuitry to
acquire operation device identification information of an operation device connected to a local area network to which the terminal is connected, and control target device identification information of a control target device connected to the local area network, and
generate and transmit the search condition information including at least one of the acquired operation device identification information or the acquired control target device identification information.

9. An operation device management apparatus, comprising:
a control record storage to store, in a manner associated with one another, specific control information indicating specific control performed on at least one control target device through an operation performed previously on an operation device, operation device identification information identifying the operation device operated, and control target device identification information identifying the at least one control target device on which the specific control has been performed, and
processing circuitry to
acquire search condition information transmitted from a terminal, the search condition information indicating a search condition for searching for at least one of control target device identification information of a control target device controllable with an operation device operated by a user, operation device identification information of an operation device to control a control target device when operated by a user, or specific control information indicating specific control having a record of being performed on the control target device,
a search the control target device identification information, the operation device identification information, and the specific control information stored in the control record storage for at least one of the control target device identification information, the operation device identification information, or the specific control information based on the search condition information, and
generate, when at least one of the control target device identification information, the operation device identification information, or the specific control information is detected, search result information including at least one of the detected control target device identification information, the detected operation device identification information, or the detected specific control information, and transmit the search result information to the terminal, wherein
the control record storage further stores, in a manner associated with the control target device identification information, class information indicating a class of the at least one control target device classified based on a type of the at least one control target device, and
when searching for at least one of the operation device identification information or the control target device identification information based on the search condition and not detecting the operation device identification information or the control target device identification information, the processing circuitry searches for at least one of operation device identification information of an operation device or control target device identification information of a control target device in a same class as at least one of the operation device being a search target or the control target device being a search target.

10. An operation device management method, comprising:
causing, by a control device, a control record storage to store, in a manner associated with one another, specific control information indicating specific control performed on at least one control target device through an operation performed previously on an operation device, operation device identification information identifying the operation device operated, and control target device identification information identifying the at least one control target device on which the specific control has been performed;
acquiring, by the control device, search condition information transmitted from a terminal, the search condition information indicating a search condition for searching for at least one of control target device identification information of a control target device controllable with an operation device operated by a user, operation device identification information of an operation device to control a control target device when operated by a user, or specific control information indicating specific control having a record of being performed on the control target device;
searching, by the control device, the control target device identification information, the operation device identification information, and the specific control information stored in the control record storage for at least one of the control target device identification information, the operation device identification information, or the specific control information based on the search condition information; and
generating, by the control device, when at least one of the control target device identification information, the operation device identification information, or the specific control information is detected, search result information including at least one of the detected control target device identification information, the detected operation device identification information, or the detected specific control information, and transmitting, by the control device, the search result information to the terminal, wherein
the control record storage further stores, in a manner associated with the control target device identification information, class information indicating a class of the at least one control target device classified based on a type of the at least one control target device, and
when searching for at least one of the operation device identification information or the control target device identification information based on the search condition and not detecting the operation device identification information or the control target device identification information, the control device searches for at least one of operation device identification information of an operation device or control target device identification information of a control target device in a same

class as at least one of the operation device being a search target or the control target device being a search target.

11. A non-transitory recording medium storing a program, the program causing a computer to function as:

a control record storage to store, in a manner associated with one another, specific control information indicating specific control performed on at least one control target device through an operation performed previously on an operation device, operation device identification information identifying the operation device operated, and control target device identification information identifying the at least one control target device on which the specific control has been performed; and

processing circuitry to

acquire search condition information transmitted from a terminal, the search condition information indicating a search condition for searching for at least one of control target device identification information of a control target device controllable with an operation device operated by a user, operation device identification information of an operation device to control a control target device when operated by a user, or specific control information indicating specific control having a record of being performed on the control target device,

search the control target device identification information, the operation device identification information, and the specific control information stored in the control record storage for at least one of the control target device identification information, the operation device identification information, or the specific control information based on the search condition information, and

generate, when at least one of the control target device identification information, the operation device identification information, or the specific control information is detected, search result information including at least one of the detected control target device identification information, the detected operation device identification information, or the detected specific control information, and transmit the search result information to the terminal, wherein

the control record storage further stores, in a manner associated with the control target device identification information, class information indicating a class of the at least one control target device classified based on a type of the at least one control target device, and

when searching for at least one of the operation device identification information or the control target device identification information based on the search condition and not detecting the operation device identification information or the control target device identification information, the processing circuitry searches for at least one of operation device identification information of an operation device or control target device identification information of a control target device in a same class as at least one of the operation device being a search target or the control target device being a search target.

12. An operation device management system, comprising:

a terminal; and

a control device, wherein

the control device includes

a control record storage to store, in a manner associated with one another, specific control information indicating specific control performed on at least one control target device through an operation performed previously on an operation device, operation device identification information identifying the operation device operated, and control target device identification information identifying the at least one control target device on which the specific control has been performed, and

first processing circuitry to

acquire search condition information transmitted from the terminal, the search condition information indicating a search condition for searching for at least one of control target device identification information of a control target device controllable with an operation device operated by a user, operation device identification information of an operation device to control a control target device when operated by a user, or specific control information indicating specific control having a record of being performed on the control target device,

search the control target device identification information, the operation device identification information, and the specific control information stored in the control record storage for at least one of the control target device identification information, the operation device identification information, or the specific control information based on the search condition information, and

generate, when at least one of the control target device identification information, the operation device identification information, or the specific control information is detected, search result information including at least one of the detected control target device identification information, the detected operation device identification information, or the detected specific control information, and transmit the search result information to the terminal, and

when the control target device identification information or the operation device identification information is not detected, the first processing circuitry generates search result information including uniform resource locator information indicating an uniform resource locator of a website showing a list of control target device identification information of a control target device recommended by a manufacturer of the operation device operated by the user or an uniform resource locator of a website showing a list of operation device identification information recommended by a manufacturer of the control target device operated by the user, and transmits the search result information to the terminal.

13. An operation device management apparatus, comprising:

a control record storage to store, in a manner associated with one another, specific control information indicating specific control performed on at least one control target device through an operation performed previously on an operation device, operation device identification information identifying the operation device operated, and control target device identification information identifying the at least one control target device on which the specific control has been performed; and

processing circuitry to

acquire search condition information transmitted from a terminal, the search condition information indicating a search condition for searching for at least one of control target device identification information of a control target device controllable with an operation device operated by a user, operation device identification information of an operation device to control a control target device when operated by a user, or specific control information indicating specific control having a record of being performed on the control target device,

search the control target device identification information, the operation device identification information, and the specific control information stored in the control record storage for at least one of the control target device identification information, the operation device identification information, or the specific control information based on the search condition information, and

generate, when at least one of the control target device identification information, the operation device identification information, or the specific control information is detected, search result information including at least one of the detected control target device identification information, the detected operation device identification information, or the detected specific control information, and transmit the search result information to the terminal, wherein

when the control target device identification information or the operation device identification information is not detected, the processing circuitry generates search result information including uniform resource locator information indicating an uniform resource locator of a website showing a list of control target device identification information of a control target device recommended by a manufacturer of the operation device operated by the user or an information recommended by a manufacturer of the control target device operated by the user, and transmits the search result information to the terminal.

14. An operation device management method, comprising:

causing, by a control device, a control record storage to store, in a manner associated with one another, specific control information indicating specific control performed on at least one control target device through an operation performed previously on an operation device, operation device identification information identifying the operation device operated, and control target device identification information identifying the at least one control target device on which the specific control has been performed;

acquiring, by the control device, search condition information transmitted from a terminal, the search condition information indicating a search condition for searching for at least one of control target device identification information of a control target device controllable with an operation device operated by a user, operation device identification information of an operation device to control a control target device when operated by a user, or specific control information indicating specific control having a record of being performed on the control target device;

searching, by the control device, the control target device identification information, the operation device identification information, and the specific control information stored in the control record storage for at least one of the control target device identification information, the operation device identification information, or the specific control information based on the search condition information;

generating, by the control device, when at least one of the control target device identification information, the operation device identification information, or the specific control information is detected, search result information including at least one of the detected control target device identification information, the detected operation device identification information, or the detected specific control information, and transmitting the search result information to the terminal; and

generating, by the control device, when the control target device identification information or the operation device identification information is not detected, search result information including uniform resource locator information indicating an uniform resource locator of a website showing a list of control target device identification information of a control target device recommended by a manufacturer of the operation device operated by the user or an uniform resource locator of a website showing a list of operation device identification information recommended by a manufacturer of the control target device operated by the user, and transmitting, by the control device, the search result information to the terminal.

15. A non-transitory recording medium storing a program, the program causing a computer to function as:

a control record storage to store, in a manner associated with one another, specific control information indicating specific control performed on at least one control target device through an operation performed previously on an operation device, operation device identification information identifying the operation device operated, and control target device identification information identifying the at least one control target device on which the specific control has been performed; and

processing circuitry to

acquire search condition information transmitted from a terminal, the search condition information indicating a search condition for searching for at least one of control target device identification information of a control target device controllable with an operation device operated by a user, operation device identification information of an operation device to control a control target device when operated by a user, or specific control information indicating specific control having a record of being performed on the control target device,

search the control target device identification information, the operation device identification information, and the specific control information stored in the control record storage for at least one of the control target device identification information, the operation device identification information, or the specific control information based on the search condition information, and

generate, when at least one of the control target device identification information, the operation device identification information, or the specific control information is detected, search result information including at least one of the detected control target device identification information, the detected operation device identification information, or the detected specific control information, and transmit the search result information to the terminal, wherein

when the control target device identification information or the operation device identification information is not detected, the processing circuitry generates search result information including uniform resource locator information indicating an uniform resource locator of a website showing a list of control target device identification information of a control target device recommended by a manufacturer of the operation device operated by the user or an information recommended by a manufacturer of the control target device operated by the user, and transmits the search result information to the terminal.

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